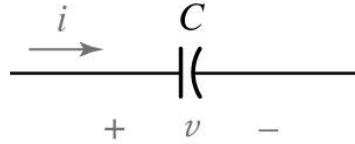


Capacitors and Inductors

Chapter 7

Ideal Capacitor Model



$$i = C \frac{dv}{dt}$$

- 1) There is no current through capacitor if $V = \text{const}$ (open circuit to DC)
- 2) Capacitor resists an abrupt change in the voltage across it
- 3) Capacitor never dissipates energy, but only stores it

Capacitor consists of 2 conducting surfaces on which charge may be stored, separated by thin insulating layer

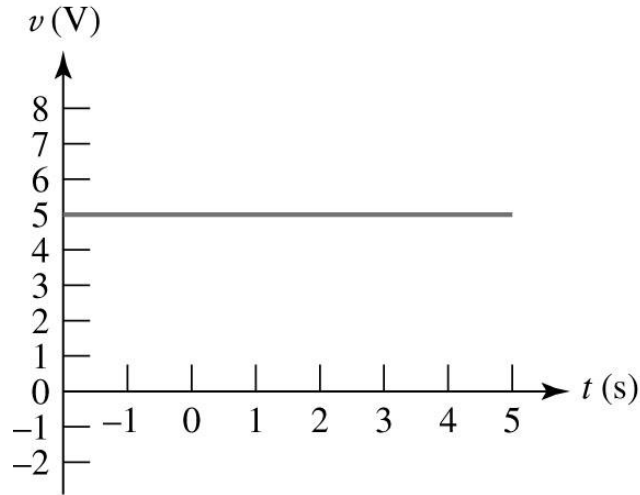
$$C = \frac{\epsilon A}{d}$$

For air or vacuum permittivity $\epsilon = \epsilon_0 = 8.854 \text{ pF/m}$

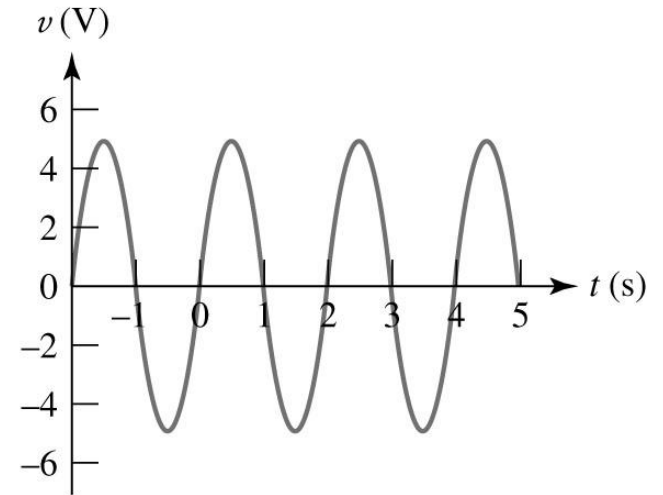
$$F = \frac{A \cdot s}{V} = \frac{J}{V^2} = \frac{W \cdot s}{V^2} = \frac{C}{V} = \frac{C^2}{J} = \frac{C^2}{N \cdot m} = \frac{s^2 \cdot C^2}{m^2 \cdot kg} = \frac{s^4 \cdot A^2}{m^2 \cdot kg} = \frac{s}{\Omega}$$

Ideal Capacitor Model

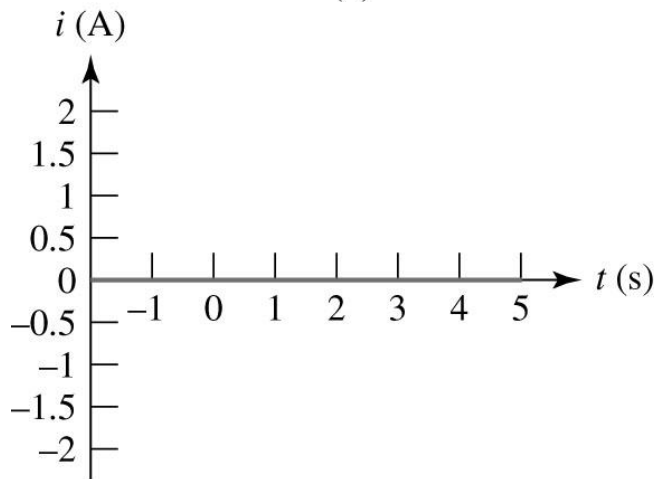
$$i = C \frac{dv}{dt}$$



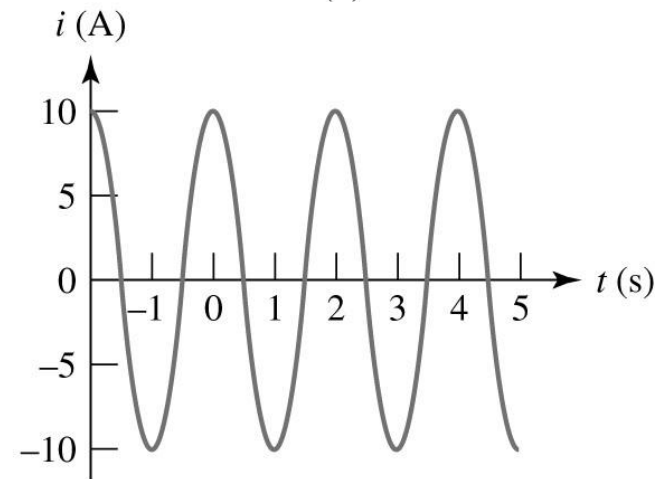
(a)



(b)



(a)



(b)

The current in the capacitor leads the voltage

Integral Voltage Current Relationships

$$dv = \frac{1}{C} i(t) dt$$

$$v(t) = \frac{1}{C} \int_{t_0}^t i(t') dt' + v(t_0)$$

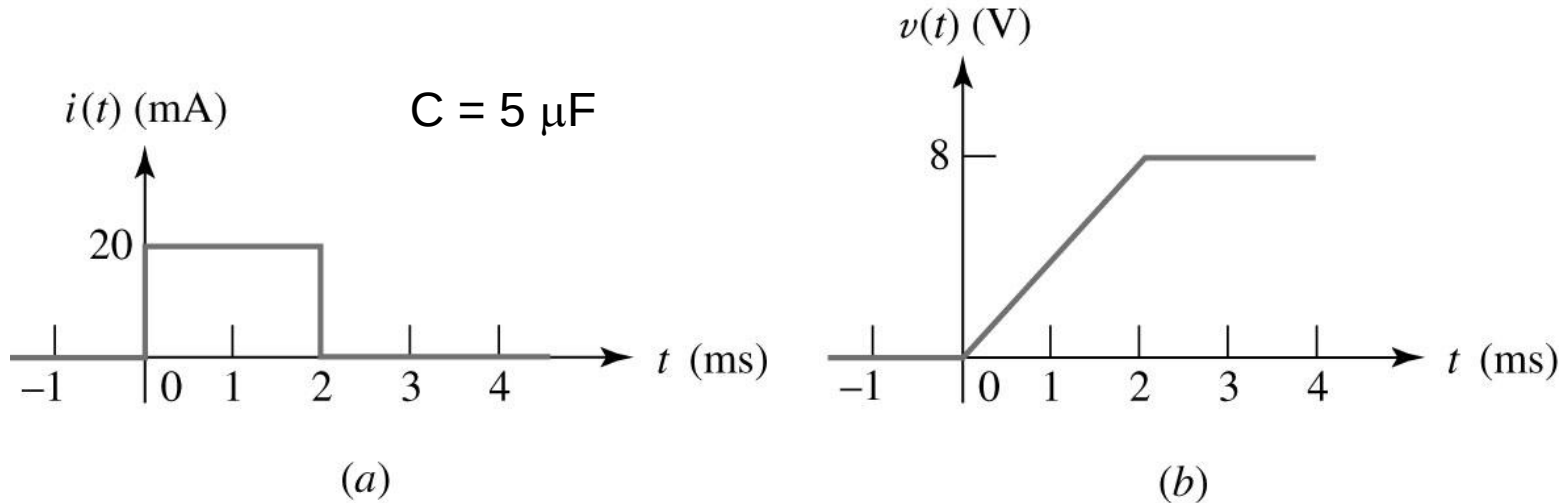
$$v(t) = \frac{1}{C} \int i dt + k$$

$$v(t) = \frac{1}{C} \int_{-\infty}^t i dt'$$

$$q(t) = Cv(t)$$

Example

Find the capacitor voltage associated with the current shown graphically



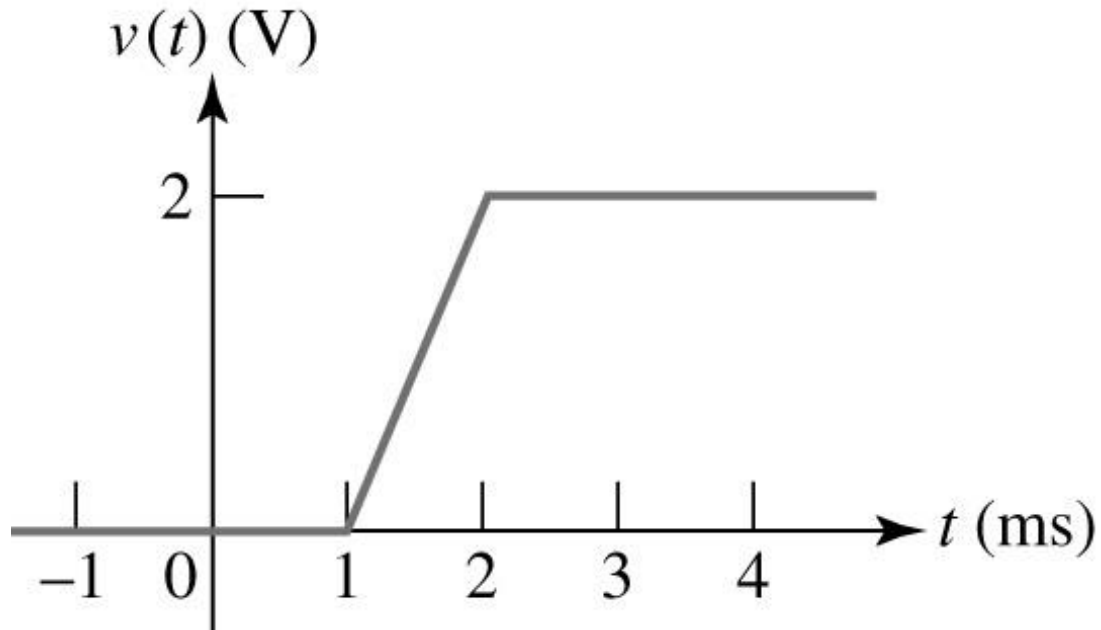
$$v(t) = 0 \text{ at } t \leq 0$$

$$v(t) = \frac{1}{5 \times 10^{-6}} \int_0^t 20 \times 10^{-3} dt' + v(0)$$

$$v(t) = 4000t \text{ at } 0 \leq t \leq 2 \text{ ms}$$

Practice

Find the current through 100 pF capacitor if its voltage as a function of time is shown graphically



Energy Storage

The power delivered to capacitor

$$p = vi = Cv \frac{dv}{dt}$$

The change in stored energy

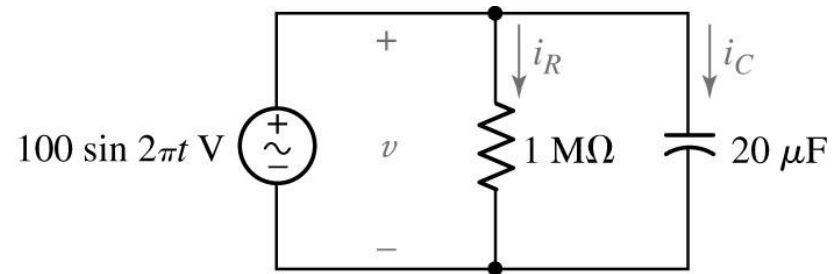
$$\int_{t_0}^t p dt' = C \int_{t_0}^t v \frac{dv}{dt'} dt' = C \int_{t_0}^t v' dv' = \frac{1}{2} C \{ [v(t)]^2 - [v(t_0)]^2 \}$$

$$w_C(t) - w_C(t_0) = \frac{1}{2} C \{ [v(t)]^2 - [v(t_0)]^2 \}$$

$$w_C(t) = \frac{1}{2} C v(t)^2$$

Example

Find the maximum energy stored in the capacitor and the energy dissipated in the resistor over the interval $0 < t < 0.5$ s



$$w_C(t) = \frac{1}{2} C v(t)^2 = 0.1 \sin^2 2\pi t$$

$$i_R = \frac{v}{R} = 10^{-4} \sin 2\pi t$$

$$p_R = i_R^2 R = (10^{-8})(10^6) \sin^2 2\pi t$$

$$w_R = \int_0^{0.5} p_R dt = \int_0^{0.5} (10^{-2}) \sin^2 2\pi t dt$$

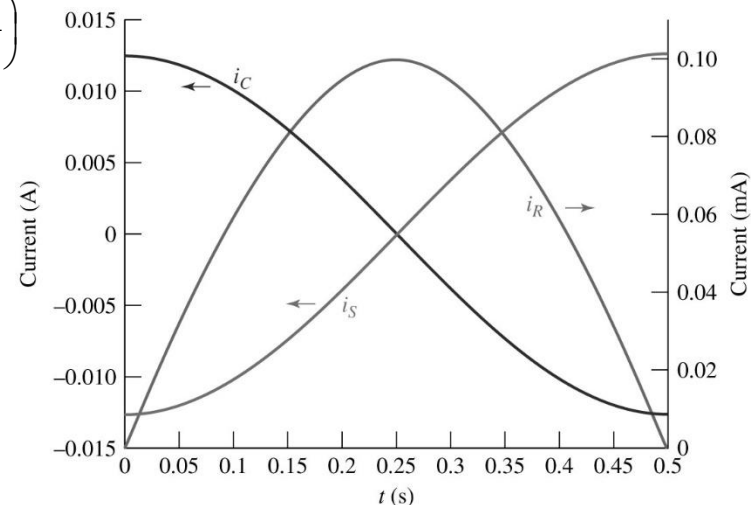
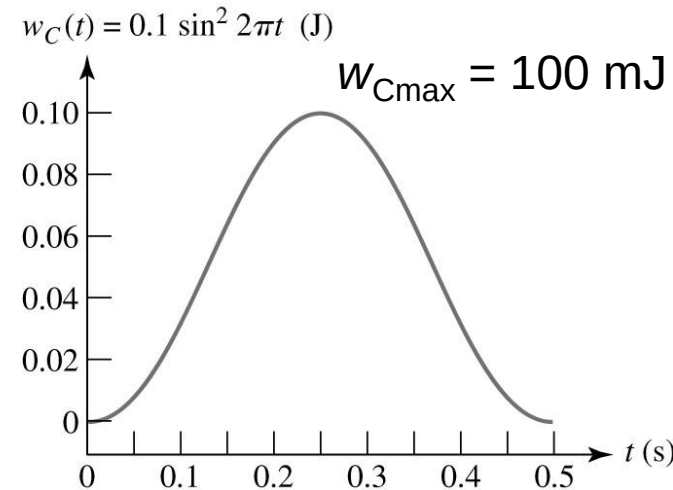
$$w_R = 2.5 \text{ mJ}$$

$$\int \sin^2 x dx = \frac{1}{2} \left(x - \frac{\sin 2x}{2} \right)$$

Capacitor current:

$$i_C = 20 \times 10^{-6} \frac{dv}{dt} = 0.004\pi \cos 2\pi t$$

Current flowing into the voltage source: $i_S = -i_C - i_R$



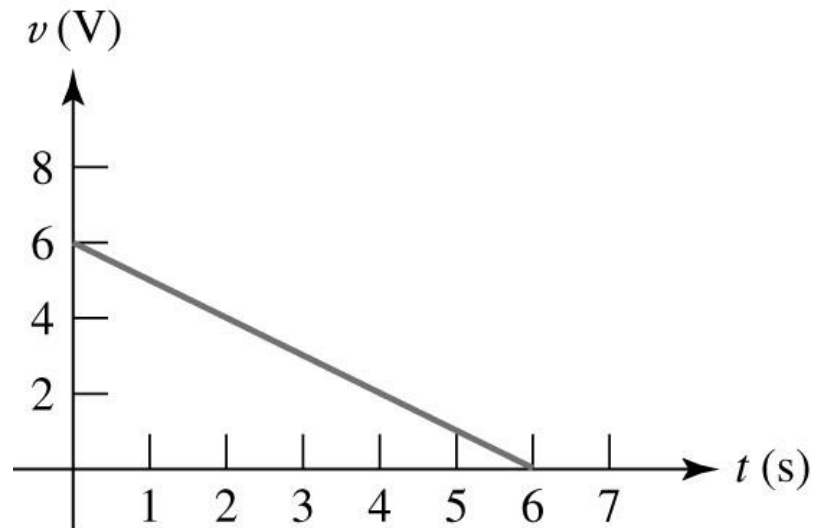
Practice

Calculate the energy stored in a $1000\ \mu\text{F}$ capacitor at $t = 50\ \mu\text{s}$ if the voltage across it is $1.5\cos 10^5 t$ Volts

90.52 μJ

Quiz

Sketch the current flowing through a $10\ \mu\text{F}$ capacitor in response to the voltage waveform below (passive sign convention applies)



Summary

1. There is no current through capacitor if $dv/dt=0$. **Capacitor is open circuit to DC**
2. A finite amount of energy can be stored in a capacitor even if the current through the capacitor is zero (at $v = \text{const}$)
3. It is impossible to change the voltage across capacitor by a finite amount in zero time, for this requires an infinite current through the capacitor.
4. Ideal capacitor stores energy but not dissipates

Ideal Inductor model

Current produces Magnetic Field

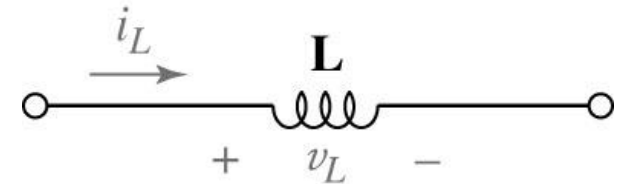
The Magnetic Field linearly depends on the Current

Changing Magnetic Field induces Voltage in neighboring circuit

The Voltage proportional to the time rate of the Current producing the Magnetic Field

$$v = L \frac{di}{dt}$$

L – Inductance



Physical inductor – coil; For coil in the form of long helix of very small pitch:

$$L = \mu N^2 A / s$$

where μ – permeability of material; N – number of complete turns of wire; A – cross-sectional area; s – axial length of the helix.

For air and vacuum: $\mu = \mu_0 = 4\pi \times 10^{-7} \text{ H/m}$

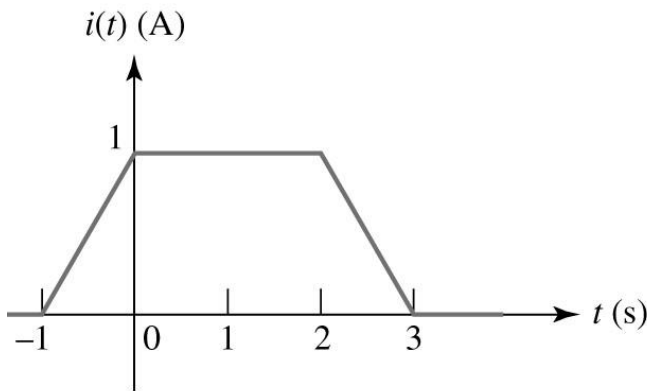
No voltage across the inductor carrying constant current

Inductor = short circuit to DC

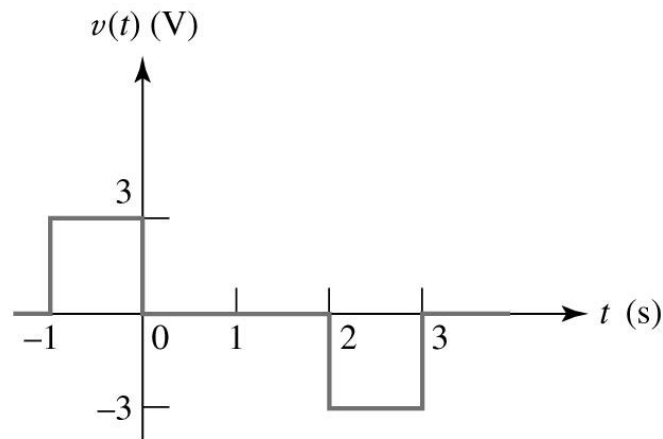
Sudden changes in inductor current requires infinite voltage

Example

Given the waveform of the current in a 3 H inductor as shown in fig. a), determine the inductor voltage and sketch it



(a)

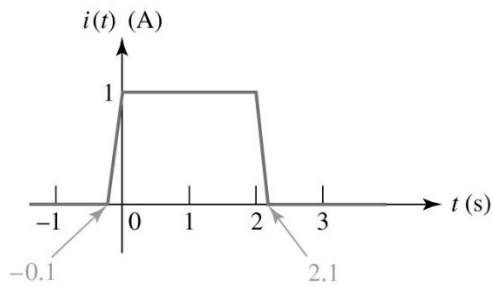


(b)

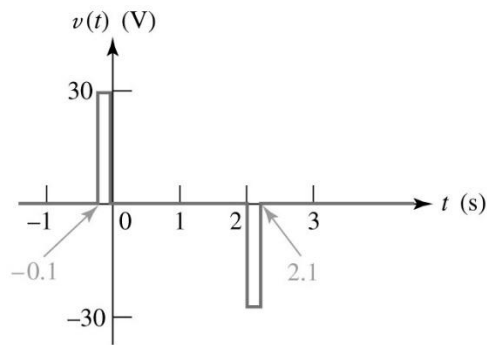
$$v = 3 \frac{di}{dt}$$

Example

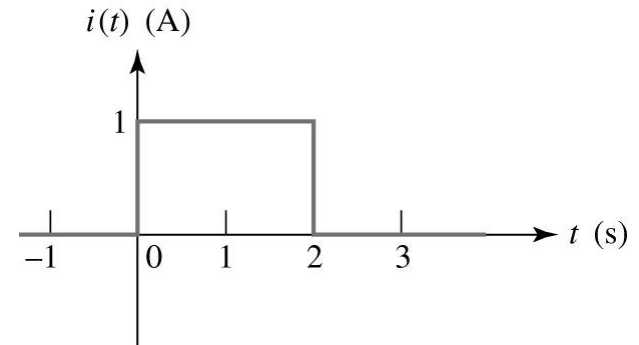
Find the inductor voltage that results from applying the current waveform shown in fig. a), to the inductor of previous example



(a)

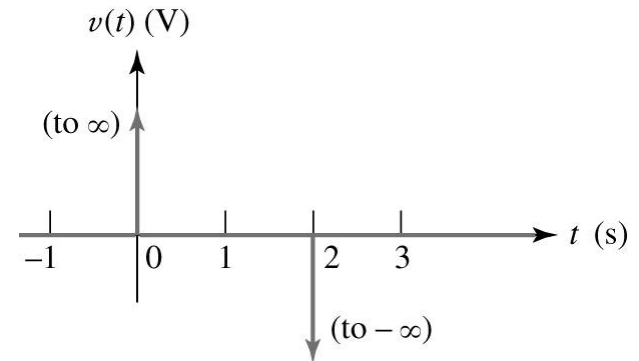


(b)



(a)

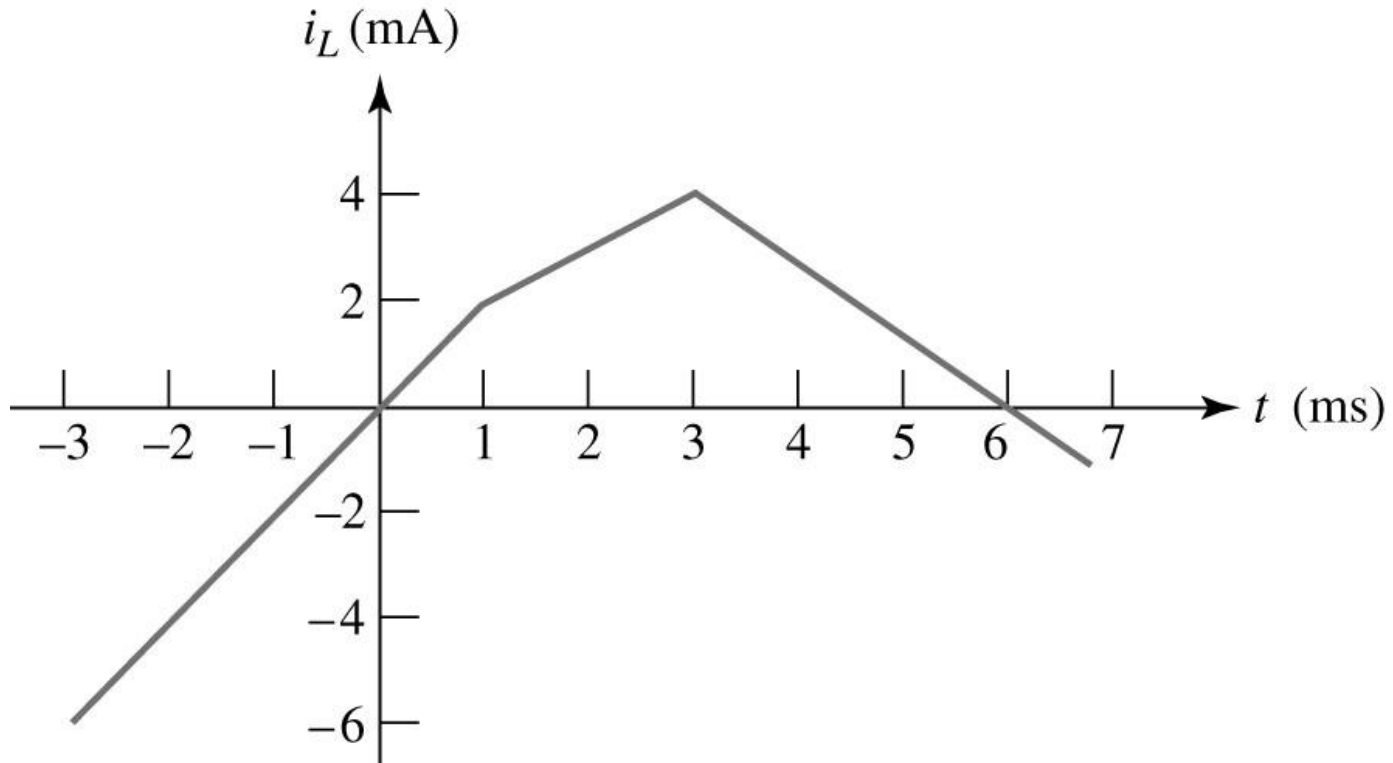
$$v = 3 \frac{di}{dt}$$



(b)

Practice

The current through 200 mH inductor is shown in the fig. Assume the passive sign convention, and find v_L at t equal to a) 0; b) 2 ms; c) 6 ms



Integral Voltage – Current Relationship

$$di = \frac{1}{L} v(t) dt$$

$$i(t) = \frac{1}{L} \int_{t_0}^t v dt' + i(t_0)$$

$$i(t) = \frac{1}{L} \int v dt + k$$

$$i(t) = \frac{1}{L} \int_{-\infty}^t v dt'$$

Example

The voltage across a 2 H inductor is known to be $6\cos 5t$ V. Determine the resulting inductor current if $i(-\pi/2) = 1$ A

$$i(t) = \frac{1}{2} \int_{t_0}^t 6 \cos 5t' dt' + i(t_0)$$

$$i(t) = \frac{1}{2} \left(\frac{6}{5} \right) \sin 5t - \frac{1}{2} \left(\frac{6}{5} \right) \sin 5t_0 + i(t_0) \quad \text{Let } t_0 = -\pi/2$$

$$i(t) = 0.6 \sin 5t - 0.6 \sin 2.5\pi + 1$$

$$i(t) = 0.6 \sin 5t + 1.6$$

Energy Storage

The power absorbed by inductor

$$p = vi = Li \frac{di}{dt}$$

The change in stored energy

$$\int_{t_0}^t p dt' = L \int_{t_0}^t i \frac{di}{dt'} dt' = L \int_{t_0}^t i di = \frac{1}{2} L \{ [i(t)]^2 - [i(t_0)]^2 \}$$

$$w_L(t) - w_L(t_0) = \frac{1}{2} L \{ [i(t)]^2 - [i(t_0)]^2 \}$$

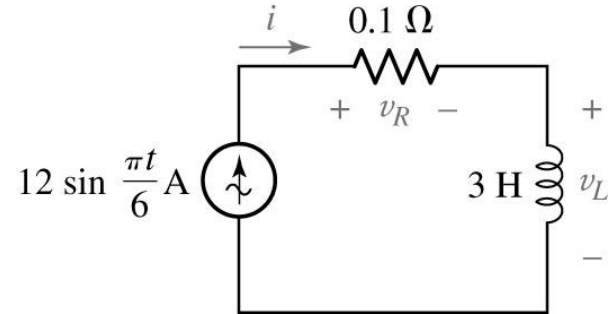
$$w_L(t) = \frac{1}{2} Li^2$$

Example

Find the maximum energy stored in the inductor and calculate how much energy is dissipated in the resistor in the time during which the energy is being stored in and then recovered from the inductor

$$w_L(t) = \frac{1}{2} L i^2 = 216 \sin^2 \frac{\pi t}{6}$$

$$\text{At } t = 3 \text{ s, } w_{L\max} = 216 \text{ J}$$



The power dissipated in the resistor:

$$p_R = i^2 R = 14.4 \sin^2 \frac{\pi t}{6}$$

The energy converted into heat during 6 s:

$$w_R = \int_0^6 p_R dt = \int_0^6 14.4 \sin^2 \frac{\pi}{6} t dt$$

$$w_R = \int_0^6 14.4 \left(\frac{1}{2} \right) \left(1 - \cos \frac{\pi}{3} t \right) dt = 43.2 \text{ J}$$

Summary

1. There is no voltage across the inductor if the current through it is not changing with time. An inductor is therefore a **short circuit to dc**.
2. A finite amount of energy can be stored in an inductor even if the voltage across the inductor is zero (at $i = \text{const}$)
3. It is impossible to change the current through an inductor by a finite amount in zero time, for this requires an infinite voltage across the inductor.
4. Ideal inductor stores energy but not dissipates

Quiz

Let $L=25$ mH for the inductor.

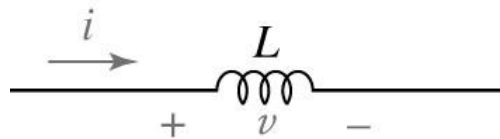
(a) Find v at $t = 12$ ms if $i = 10te^{-100t}$ A.

(b) Find i at $t = 0.1$ s if $v = 6e^{-12t}$ V and $i(0) = 10$ A.

If $i = 8(1 - e^{-40t})$ mA,

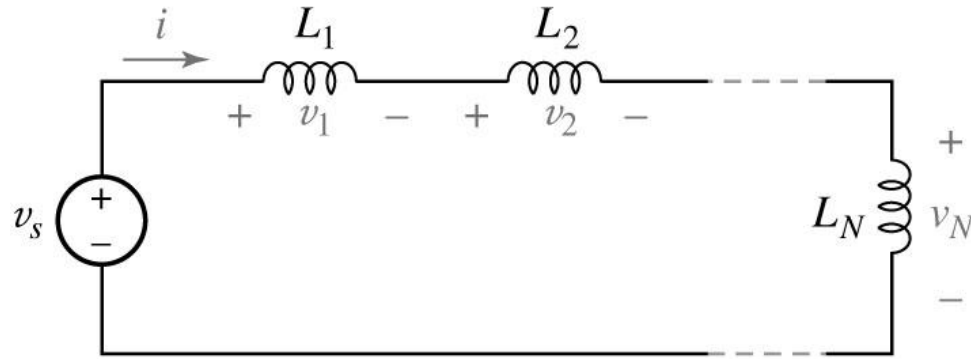
(c) Find the power being delivered to the inductor at $t = 50$ ms, and

(d) Find the energy stored in the inductor at $t = 40$ ms

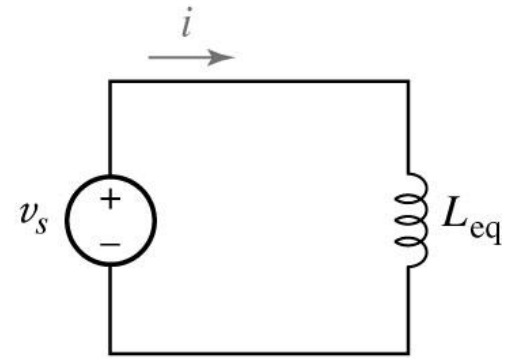


-15.06 mV; 24 A; 7.49 μ W; 0.510 μ J

Inductors in Series



(a)

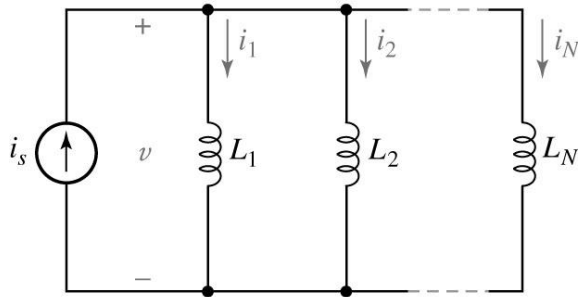


(b)

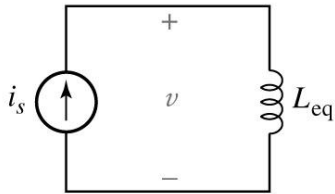
From KVL:

$$L_{eq} = \sum_{n=1}^N L_n$$

Inductors in Parallel



(a)



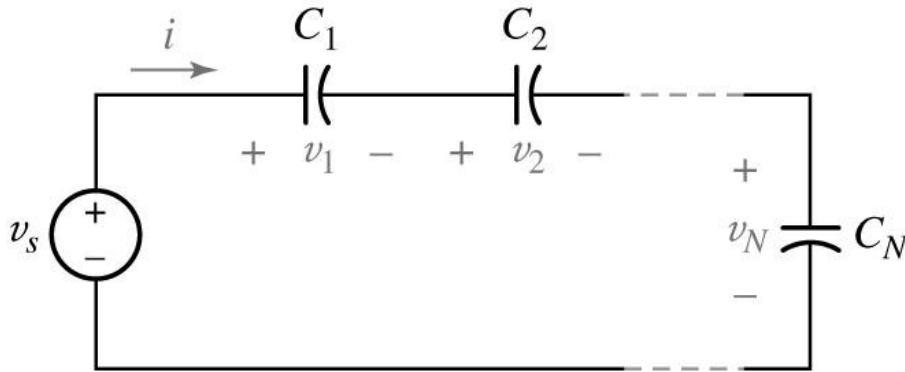
(b)

$$L_{eq} = \frac{1}{\frac{1}{L_1} + \frac{1}{L_2} + \dots + \frac{1}{L_n}}$$

For 2 inductors:

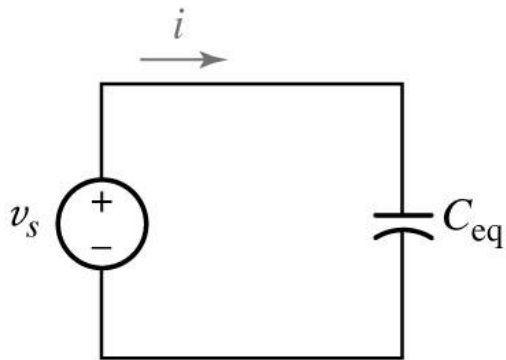
$$L_{eq} = \frac{L_1 L_2}{L_1 + L_2}$$

Capacitors in Series



(a)

$$C_{eq} = \frac{1}{\frac{1}{C_1} + \frac{1}{C_2} + \dots + \frac{1}{C_n}}$$

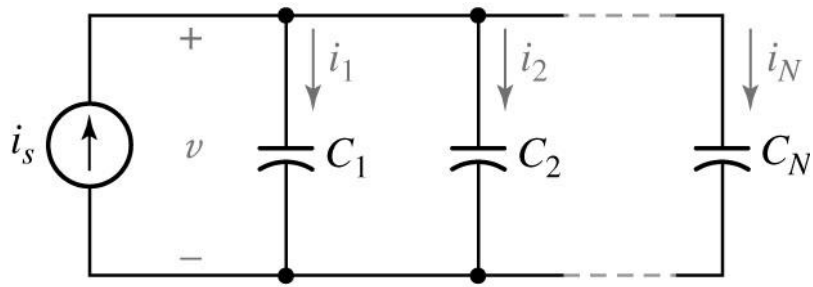


(b)

For 2 capacitors:

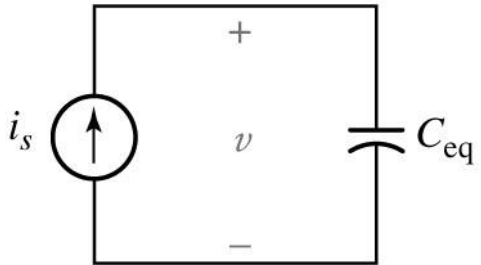
$$C_{eq} = \frac{C_1 C_2}{C_1 + C_2}$$

Capacitors in Parallel



(a)

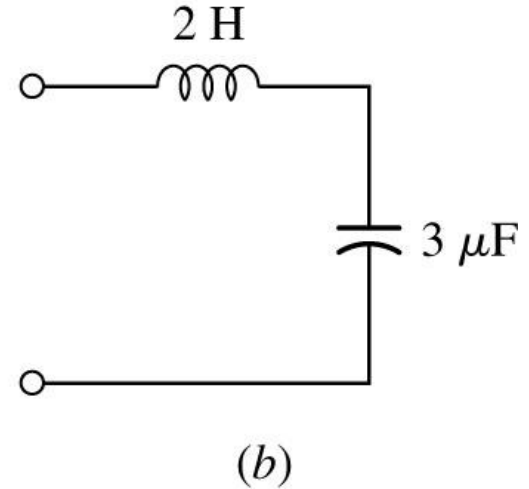
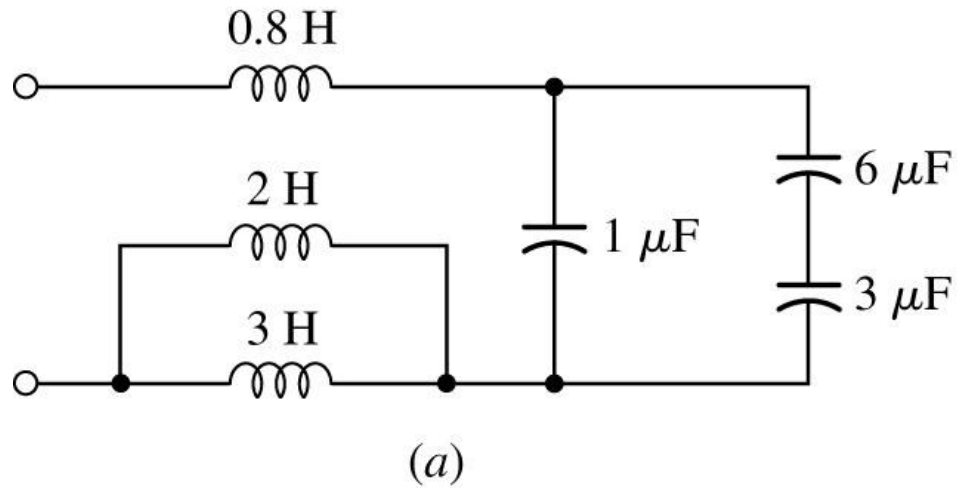
$$C_{eq} = \sum_{n=1}^N C_n$$



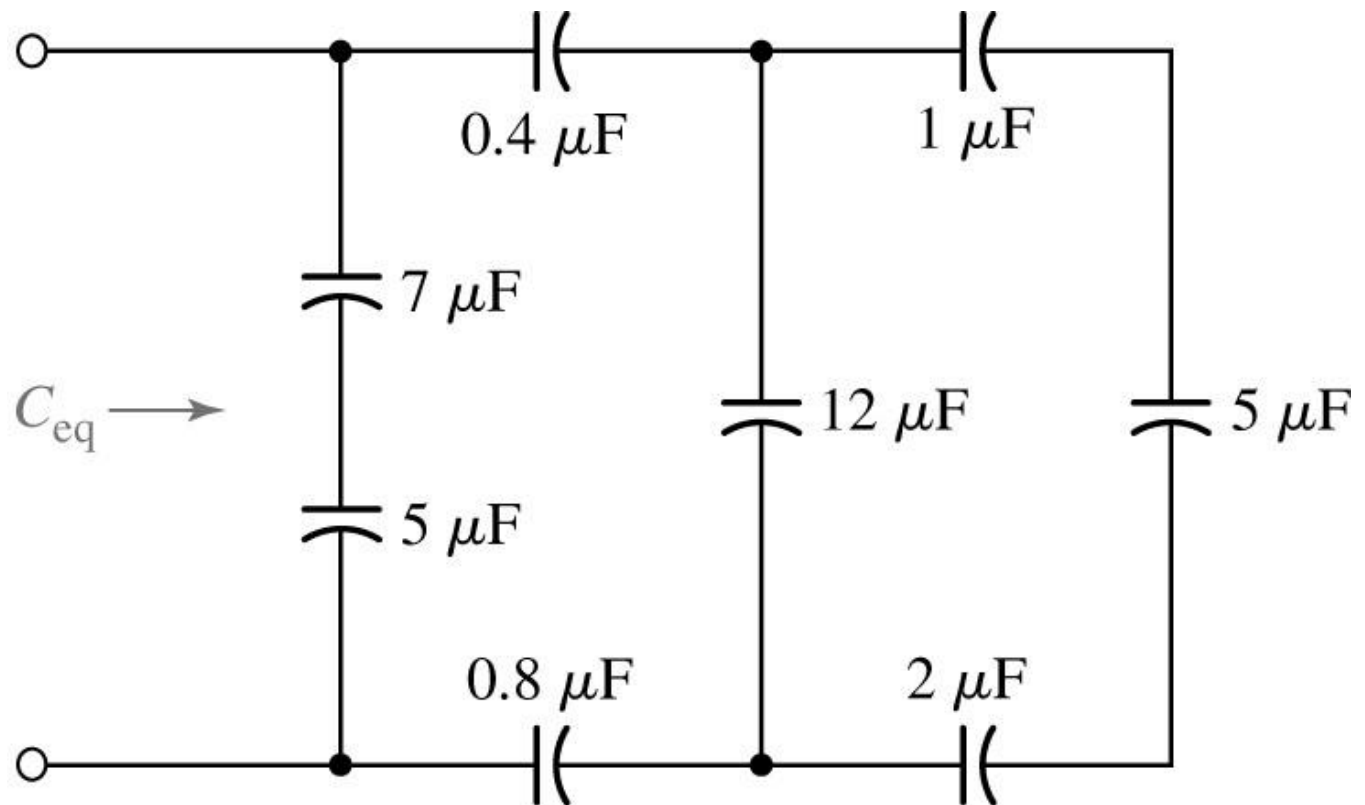
(b)

Example

Simplify the network using series/parallel combination

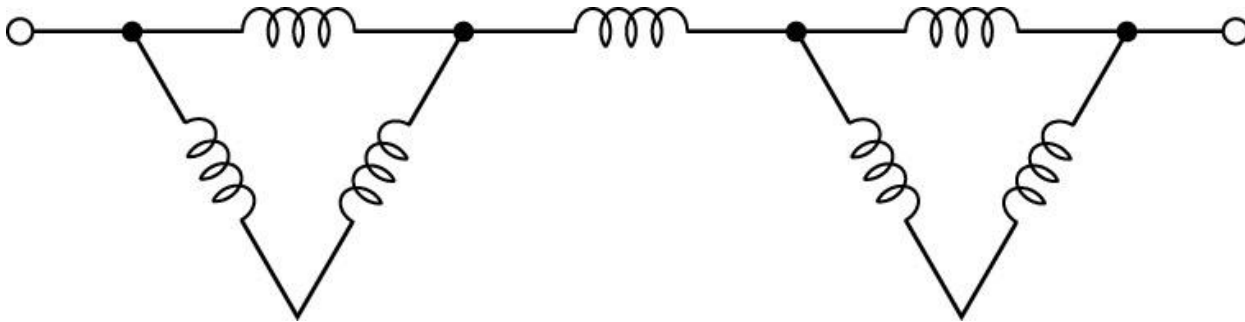
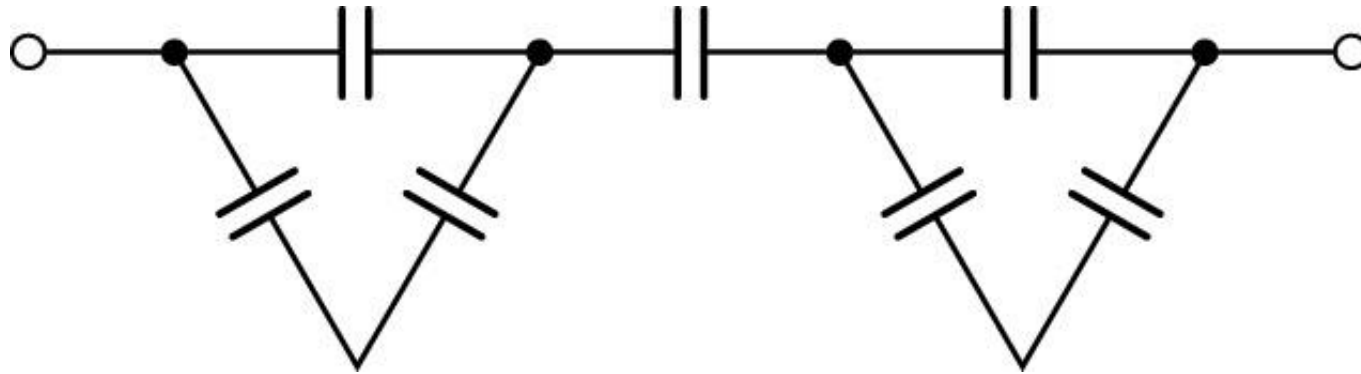


Practice

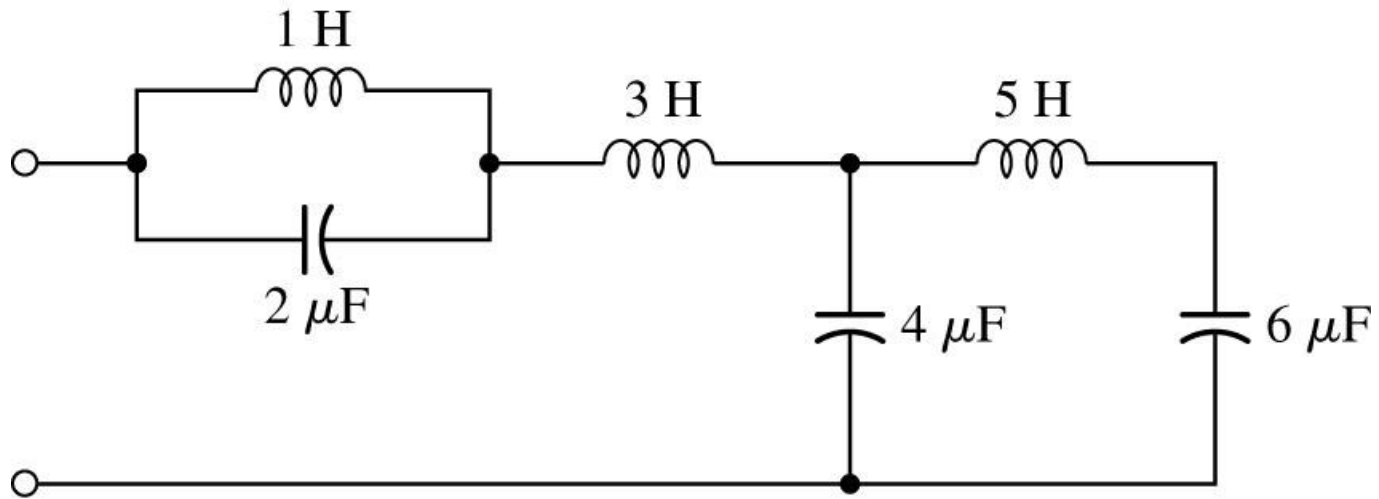


Quiz

Determine equivalent capacitance and inductors



Can we simplify it?



Consequences of Linearity

Benefits of linearity apply to RLC circuits too

Principle of superposition is applicable to RLC circuits

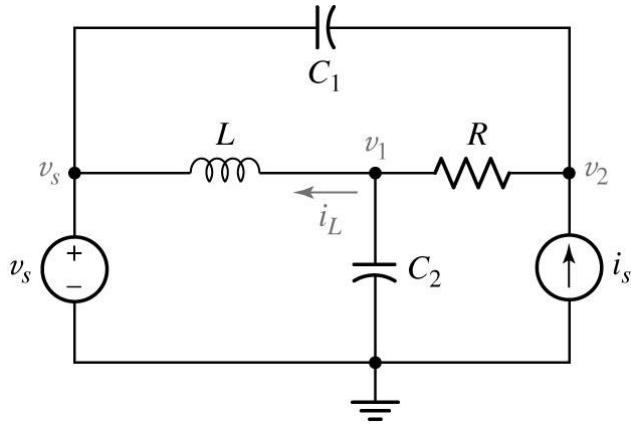
Initial inductor current appears at nodal analysis as an independent current source

Initial capacitor voltage must be treated at nodal analysis as an independent voltage source

Kirchhoff's laws are valid

Norton and Thevenin theorems are valid

Write nodal equation



The right-hand node:

$$\frac{v_1 - v_2}{R} + i_s = C_1 \frac{d(v_2 - v_s)}{dt}$$

The central node:

$$\frac{1}{L} \int_{t_0}^t (v_1 - v_s) dt' + i_L(t_0) + \frac{v_1 - v_2}{R} + C_2 \frac{dv_1}{dt} = 0$$

Is it easy to solve? Other ways?

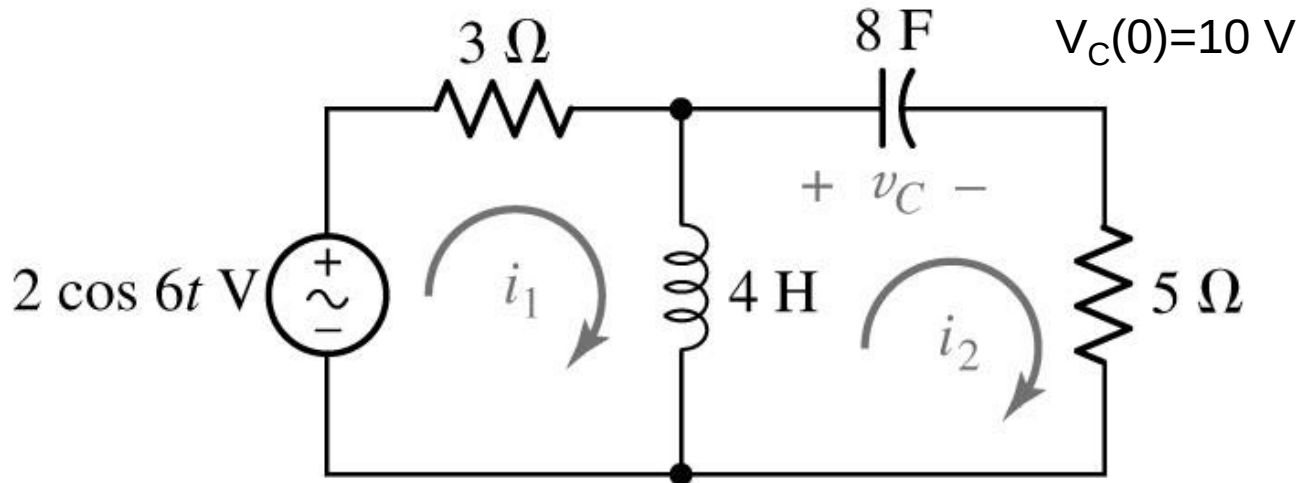
Duality

Two circuits are “**duals**” if the mesh equations that characterize one of them have the same mathematical form as the nodal equations that characterize the other

Exact duals – each mesh equation is numerically identical with the corresponding nodal equation

Duality merely refers to any of the properties exhibited by dual circuits

Non-planar circuits does not possess dual

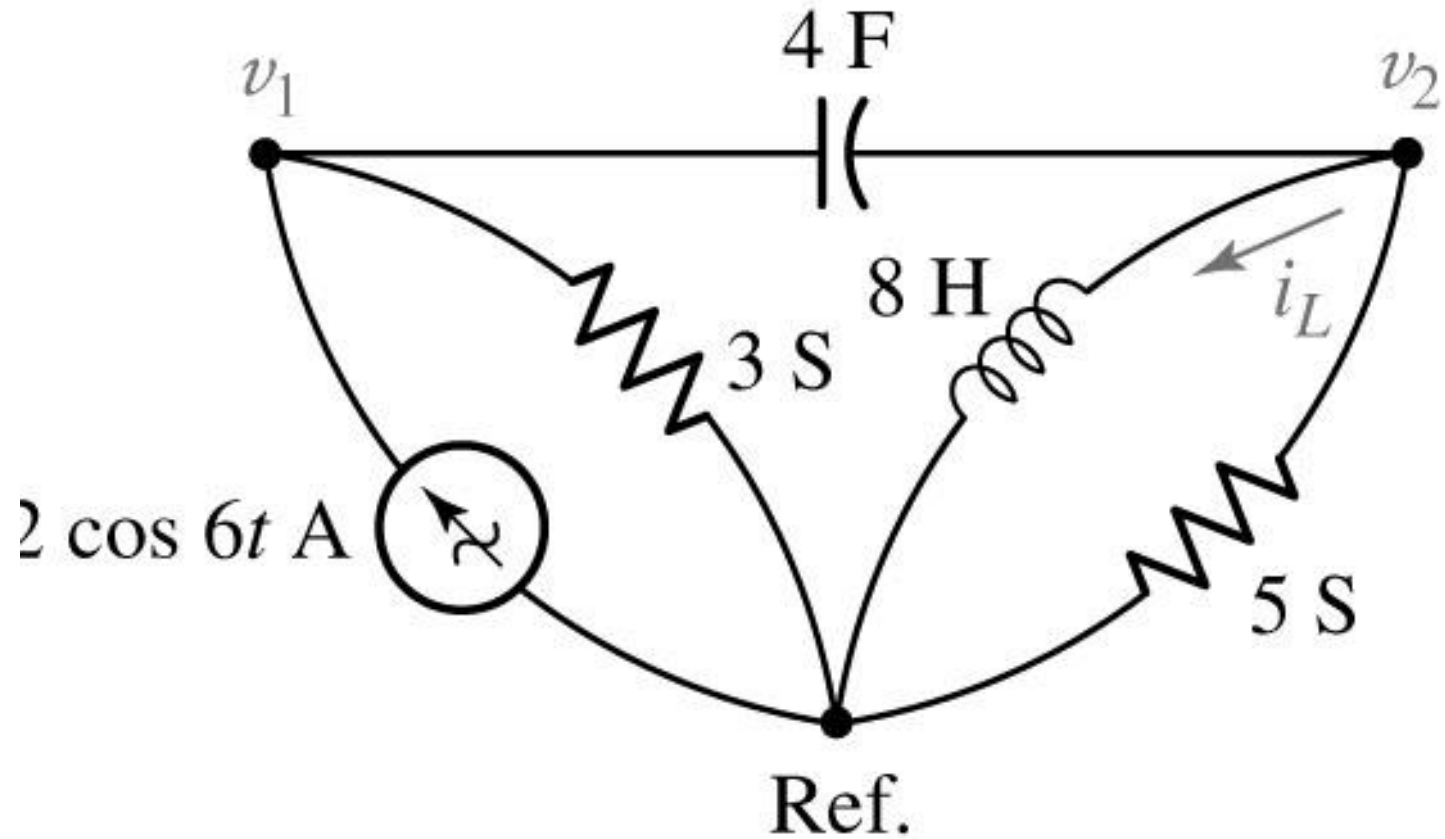


$$\begin{aligned} 3i_1 + 4 \frac{d(i_1 - i_2)}{dt} &= 2 \cos 6t \\ 4 \frac{d(i_2 - i_1)}{dt} + \frac{1}{8} \int_0^t i_2 dt' + 5i_2 &= -10 \end{aligned}$$



$$\begin{aligned} 3v_1 + 4 \frac{d(v_1 - v_2)}{dt} &= 2 \cos 6t \\ 4 \frac{d(v_2 - v_1)}{dt} + \frac{1}{8} \int_0^t v_2 dt' + 5v_2 &= -10 \end{aligned}$$

Duality



If clock-wise mesh current leaves the positive terminal of voltage source, its current source dual would have a current direction into the non-reference node; and vice versa

Duality

voltage — current

parallel — serial (circuits)

resistance — conductance

impedance — admittance

capacitance — inductance

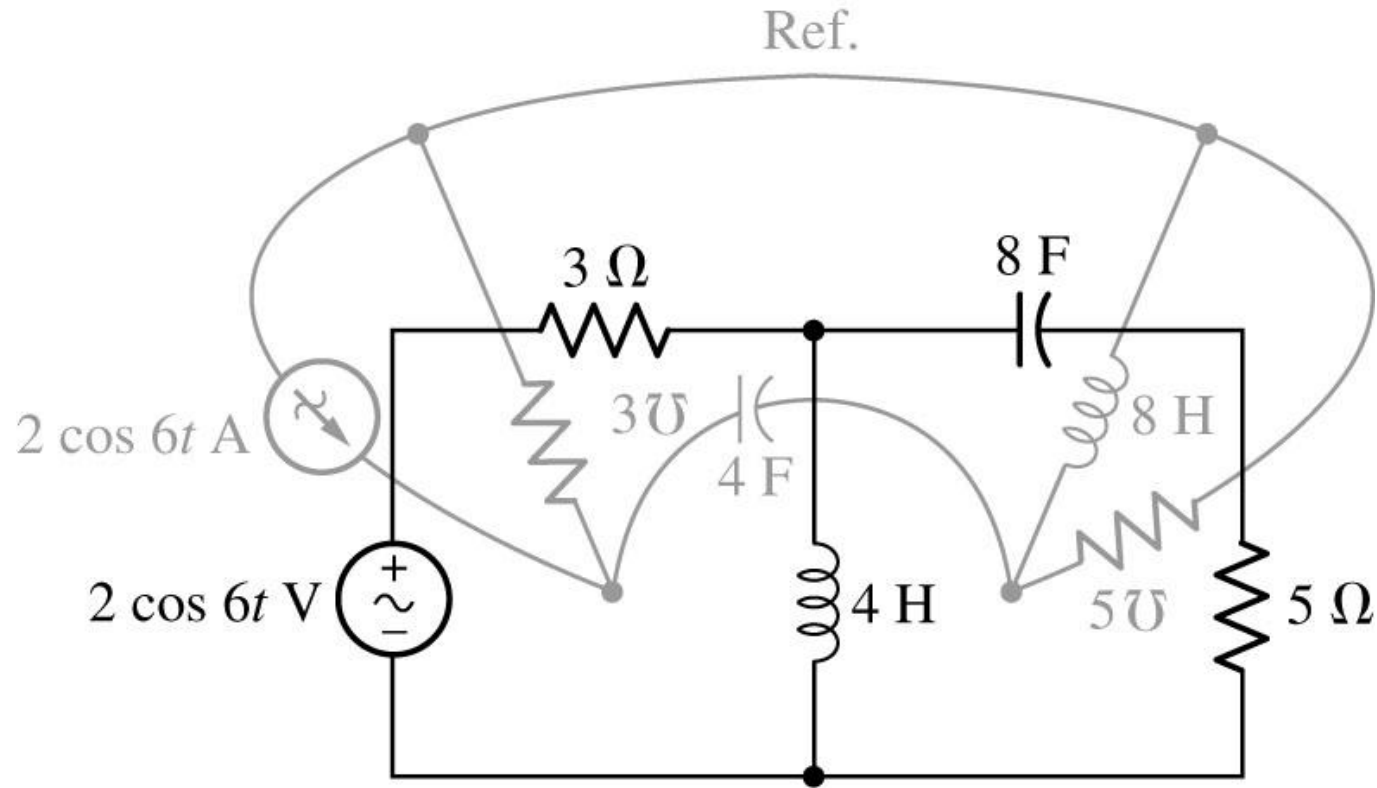
two resistances in series — two conductances in parallel

Kirchhoff's current law — Kirchhoff's voltage law

Thévenin's theorem — Norton's theorem

Elements common for two meshes transform into the elements between two nodes

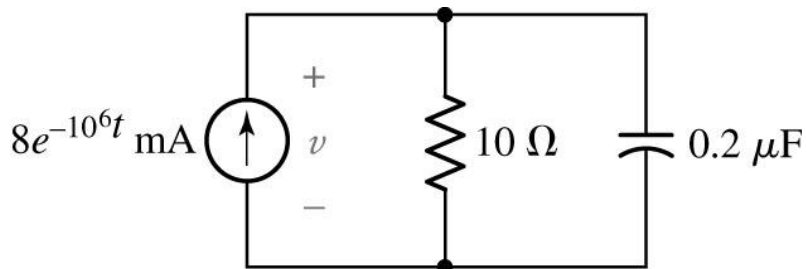
Duality



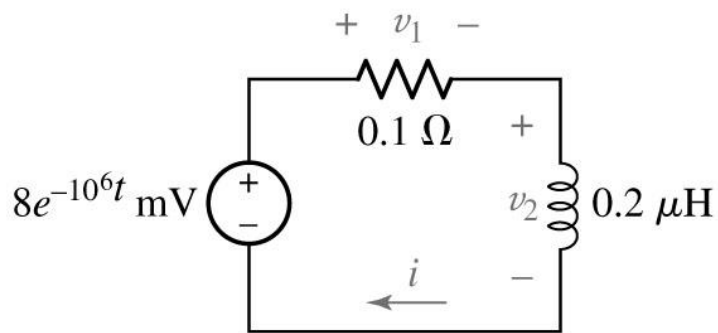
If clock-wise mesh current leaves the positive terminal of voltage source, its current source dual would have a current direction into the non-reference node; and vice versa

Practice

Write the single nodal equation for the circuit *a*) and show by direct substitution that $v = -80e^{-10^6 t}$ is a solution. Knowing this find (a) v_1 ; (b) v_2 and (c) i for the circuit *b*)



(a)



(b)

$$8 \times 10^{-3} e^{-10^6 t} = 0.1v + 0.2 \times 10^{-6} \frac{dv}{dt}$$

Invoking duality

$$v = -80e^{-10^6 t}$$

$$v_1 = 0.1v = \underline{-8e^{-10^6 t} \text{ mV}}$$

$$v_2 = 8 \times 10^{-3} e^{-10^6 t} - v_1 = \underline{16e^{-10^6 t} \text{ mV}}$$

$$i = \frac{v_1}{0.1} = \underline{-80e^{-10^6 t} \text{ mA}}$$

BASIC RL
and
RC CIRCUITS

THE SOURCE FREE RL-CIRCUITS

Homogeneous linear differential equation: every term is of the first order in the dependent variable or one of its derivatives

Solution: expression for the dependent variable that satisfies both the differential equation and also the prescribed energy distribution in the inductors or capacitors at a prescribed instant of time (initial conditions), usually at $t = 0$

Solution:

$$\text{Response} = \text{forced response} + \text{complementary response}$$

complementary response – response of the circuit if there is no source:

natural response

transient response

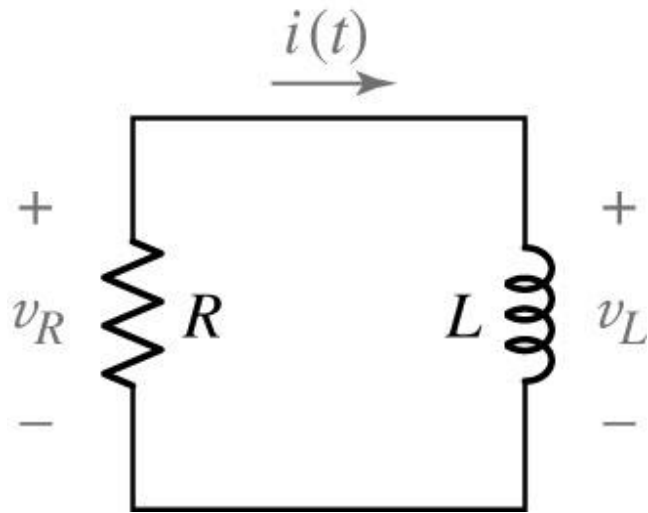
free response

forced response – if independent sources act on a circuit

particular solution

steady-state response

THE SOURCE FREE RL-CIRCUITS



$$i = I_0 \text{ at } t = 0$$

$$Ri + v_L = Ri + L \frac{di}{dt} = 0$$

$$\frac{di}{dt} + \frac{R}{L}i = 0$$

$$\frac{di}{i} = -\frac{R}{L}dt$$

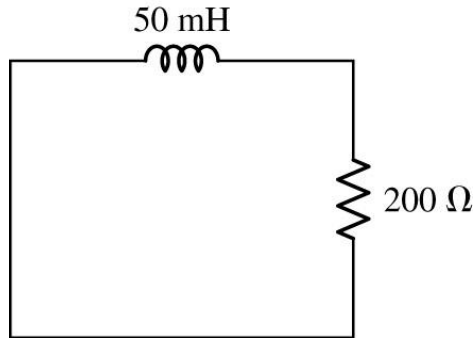
$$\int_{I_0}^{i(t)} \frac{di'}{i'} = \int_0^t \left(-\frac{R}{L}\right) dt'$$

$$\ln i \Big|_{I_0}^i = -\frac{R}{L} t' \Big|_0^t$$

$$\ln i - \ln I_0 = -\frac{R}{L}(t - 0)$$

$$i(t) = I_0 e^{-Rt/L}$$

Example



If the inductor has a current $i_L = 2$ A at $t = 0$
find an expression for $i_L(t)$ valid for $t > 0$,
and its value at $t = 200 \mu\text{s}$

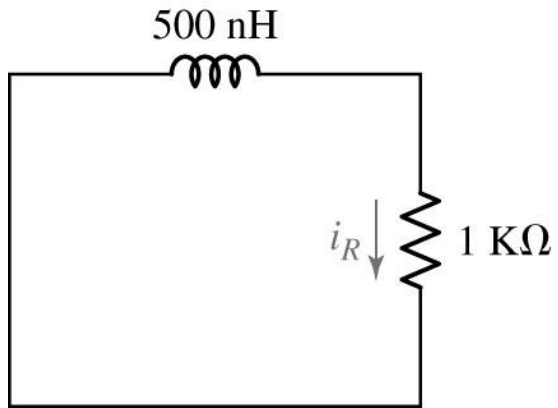
$$i_L(t) = I_0 e^{-Rt/L}$$

$$i_L(t) = 2e^{-4000t}$$

$$t = 200 \times 10^{-6} \text{ s}$$

$$i_L(t) = 898.7 \text{ mA}$$

Quiz



Determine the current i_R through the resistor at $t = 1 \text{ ns}$ if $i_R(0) = 6 \text{ A}$

General Solution Approach

Assume solution $i(t) = Ae^{s_1 t}$ $A, s_1 - \text{const}$

$$\frac{di}{dt} + \frac{R}{L}i = 0$$

$$As_1 e^{s_1 t} + A \frac{R}{L} e^{s_1 t} = 0$$

$$(s_1 + \frac{R}{L})Ae^{s_1 t} = 0 \quad \begin{cases} A = 0 \\ s_1 = -\infty \end{cases} \quad \text{All solutions equal 0}$$

$$s_1 = -\frac{R}{L}$$

$$i(t) = Ae^{-Rt/L}$$

$$i = I_0 \text{ at } t = 0 \quad A = I_0 \quad i(t) = I_0 e^{-Rt/L}$$

$$s_1 + \frac{R}{L} = 0$$

characteristic equation

Assume a general solution with appropriate constants.



Substitute the trial solution into the differential equation and simplify the result.



Determine the value for one constant that does not result in a trivial solution.



Invoke the initial condition(s) to determine values for the remaining constant(s).



End.

General Solution Approach

$$a \frac{df}{dt} + bf = 0$$

a, b – const

Substitute s^1 for df/dt and s^0 for f

$$a \frac{df}{dt} + bf = as + b = 0$$

$as + b = 0$ – characteristic equation

$$s = -\frac{b}{a}$$

Solution for our differential equation:

$$f = Ae^{-bt/a}$$

Assume a general solution with appropriate constants.



Substitute the trial solution into the differential equation and simplify the result.



Determine the value for one constant that does not result in a trivial solution.



Invoke the initial condition(s) to determine values for the remaining constant(s).



End.

Example

For the circuit a) find the voltage labeled v at $t = 200$ ms

For circuit c): $-v + 10i_L + 5\frac{di_L}{dt} = 0 \quad i_L = -v/40$

$$\frac{5}{40} \frac{dv}{dt} + \left(\frac{10}{40} + 1\right)v = 0 \quad \frac{dv}{dt} + 10v = 0$$

Characteristic equation: $s + 10 = 0; \quad s = -10$

For circuit b): $i_L = 24/10 = 2.4$ A

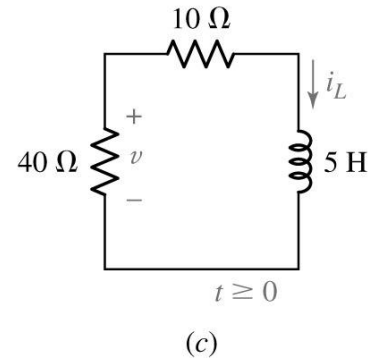
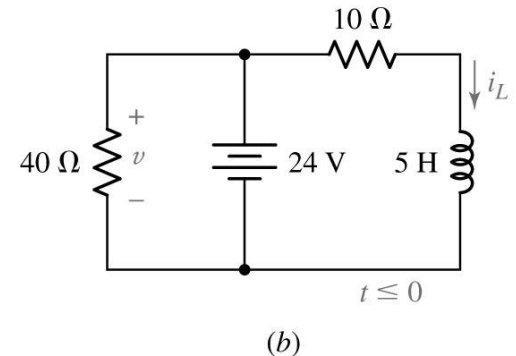
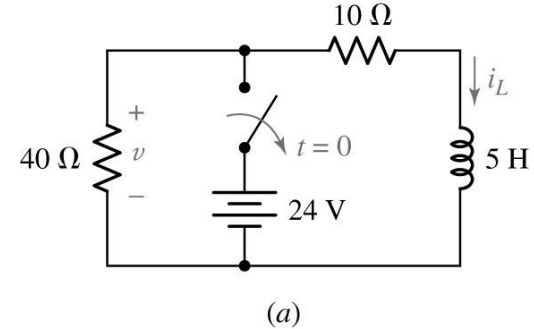
For circuit c): $v(0) = (-2.4)(40) = -96$ V

$$v(t) = Ae^{-10t} \quad v(t) = -96e^{-10t}$$

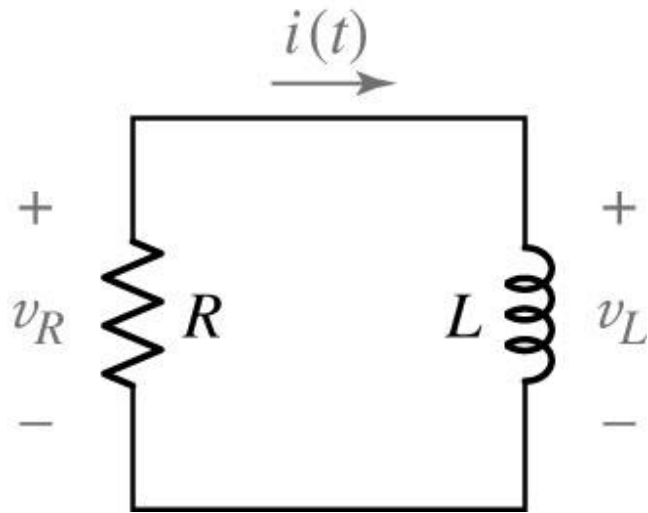
$$v(0.2) = -12.99 \text{ V}$$

We can find inductor current knowing that $v = -40i_L$

$$i_L(t) = 2.4e^{-10t}$$



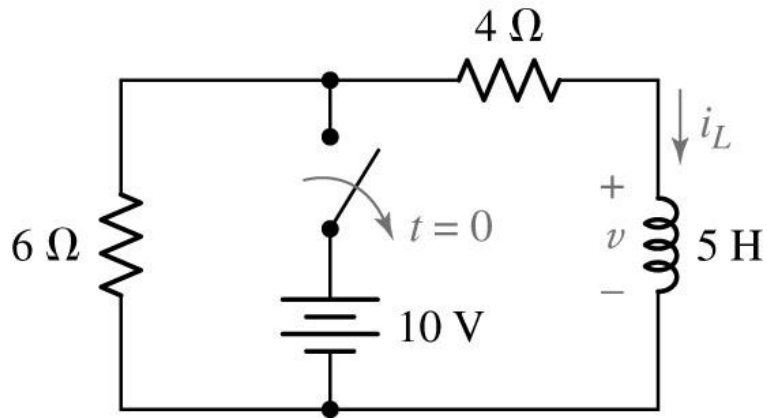
Power and Energy



$$p_R = i^2 R = I_0^2 R e^{-2Rt/L}$$

$$w_R = \int_0^{\infty} p_R dt = I_0^2 R \int_0^{\infty} e^{-2Rt/L} dt = I_0^2 R \left(-\frac{L}{2R} \right) e^{-2Rt/L} \Big|_0^{\infty} = \frac{1}{2} L I_0^2$$

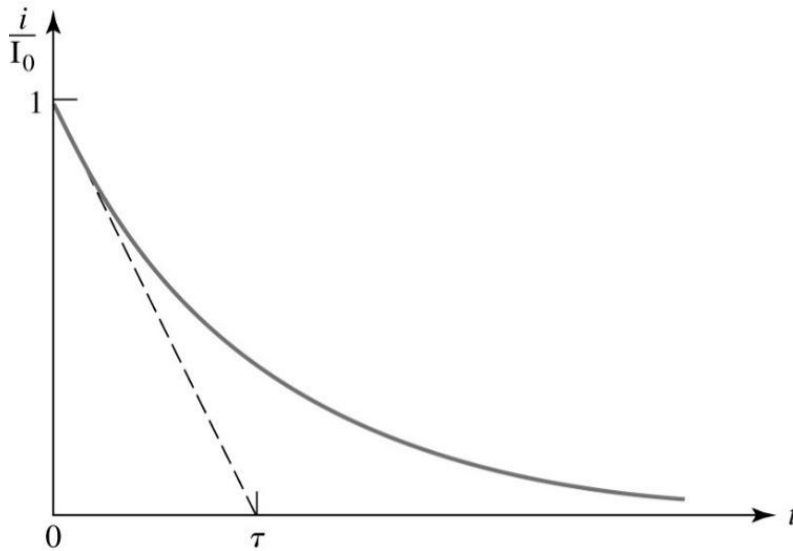
Quiz



Determine the inductor voltage v for $t > 0$

$$-25\exp(-2t) , \text{ V}$$

Inductor's Response



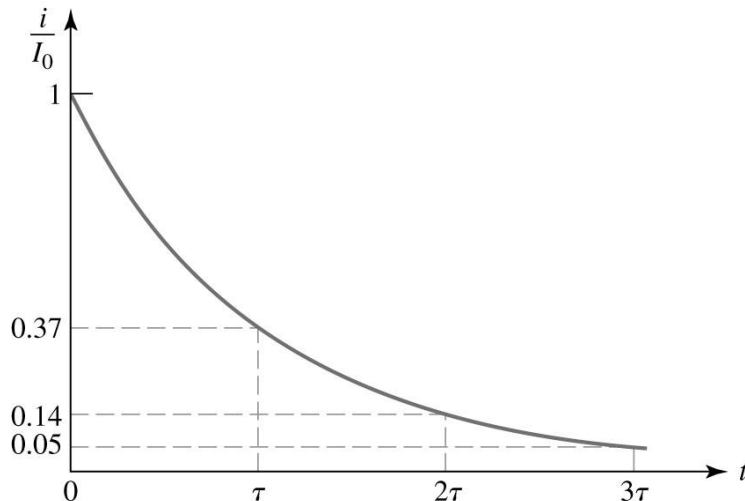
$$i(t) = I_0 e^{-Rt/L}$$

$$t = 0; i = I_0$$

The graph will not change if **R/L** = const

Initial rate of decay: $\left. \frac{d}{dt} \frac{i}{I_0} \right|_{t=0} = -\frac{R}{L} e^{-Rt/L} \bigg|_{t=0} = -\frac{R}{L}$

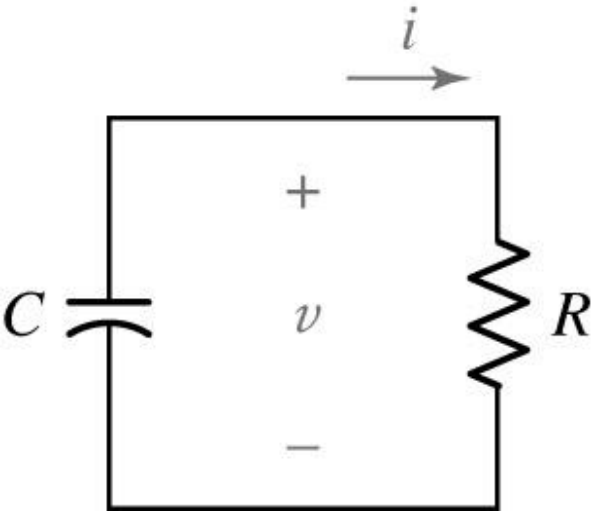
τ – the value of time it takes for I_0 to decrease by $e = 2.718$ times or to 36.8 % of initial value



$$\tau = \frac{L}{R} \quad \text{– time constant}$$

$$\frac{i(\tau)}{I_0} = e^{-1} = 0.3679$$

The source-free RC circuit



Assume initial stored energy in the capacitor $v(0) = V_0$

The currents leaving top node through each element:

$$C \frac{dv}{dt} + \frac{v}{R} = 0$$

$$\frac{dv}{dt} + \frac{v}{CR} = 0 \quad \text{similar to} \quad \frac{di}{dt} + \frac{R}{L}i = 0$$

(duality)

$$\text{If } i(t) = i(0)e^{-Rt/L} = I_0 e^{-Rt/L} \quad \Rightarrow \quad v(t) = v(0)e^{-t/RC} = V_0 e^{-t/RC}$$

We can use another way (KVL):

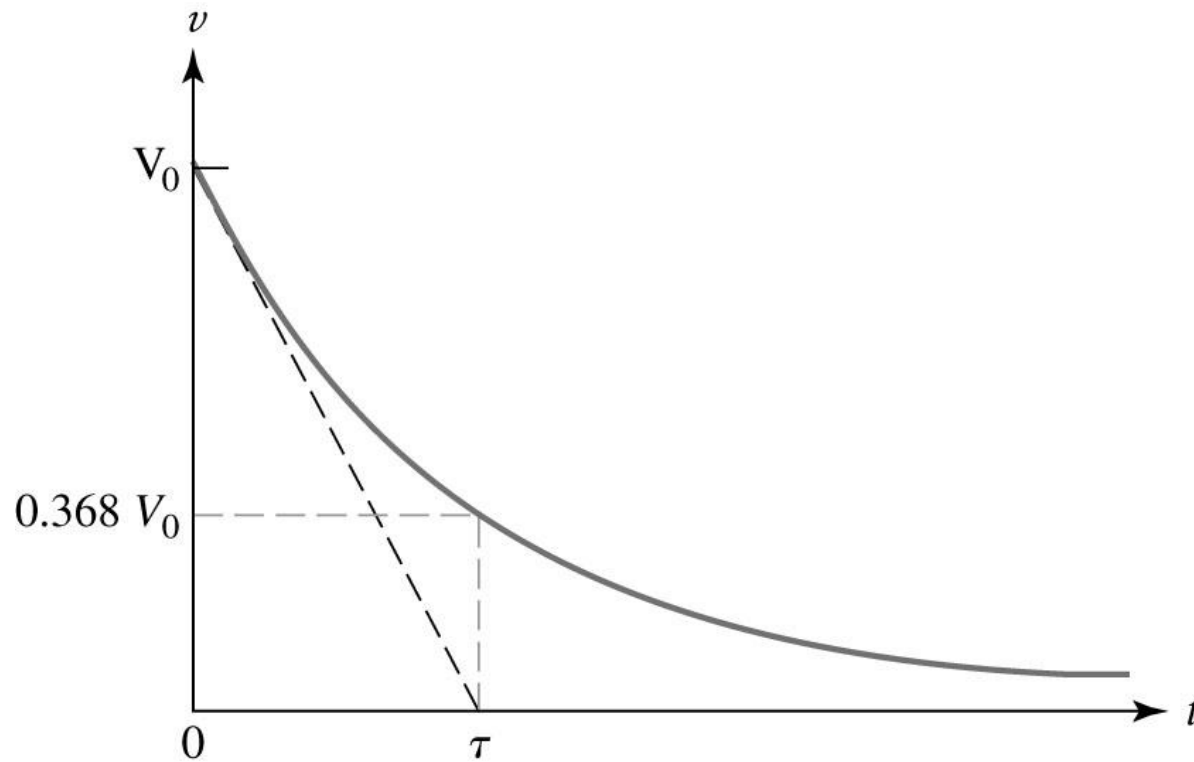
$$-v + v = -\frac{1}{C} \int_0^t (-i) dt' - V_0 + Ri = 0$$

Time derivative: $\frac{i}{C} + R \frac{di}{dt} = 0$ $i = \frac{v}{R}$ $\frac{v}{RC} + \frac{dv}{dt} = 0$ again

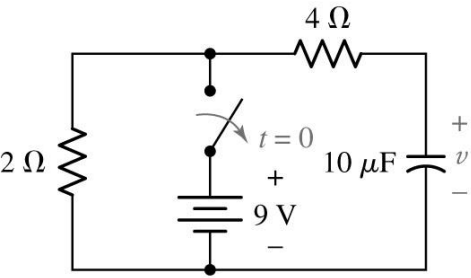
Time constant

$$\frac{1}{RC} \tau = 1$$

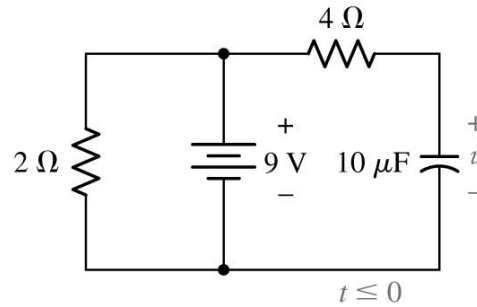
$\tau = RC$ – time constant



Example



(a)

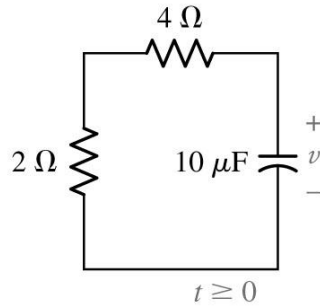


(b)

Find the voltage labeled v at $t = 200 \mu\text{s}$

$$v(0) = 9 \text{ V}$$

$$\tau = RC = (2 + 4)(10 \times 10^{-6}) = 60 \times 10^{-6} \text{ s}$$



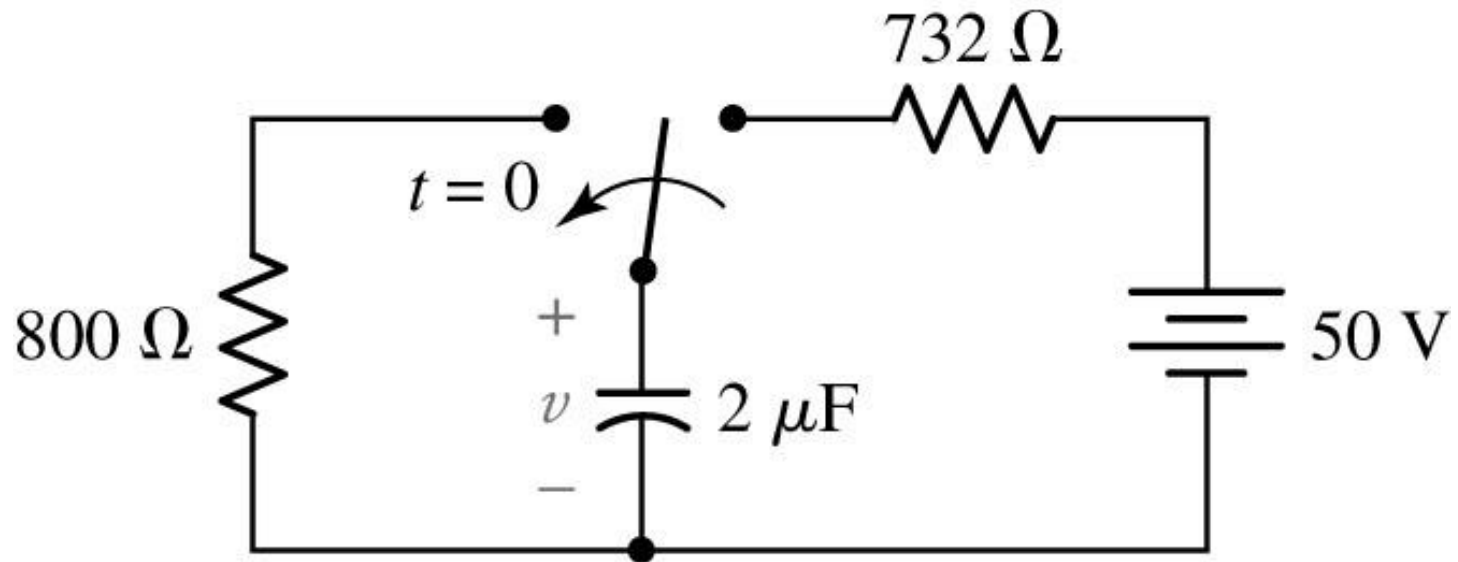
(c)

$$v(t) = v(0)e^{-t/RC} = v(0)e^{-t/60 \cdot 10^{-6}} = 9e^{-t/60 \cdot 10^{-6}}$$

$$v(200 \times 10^{-6}) = 321.1 \text{ mV}$$

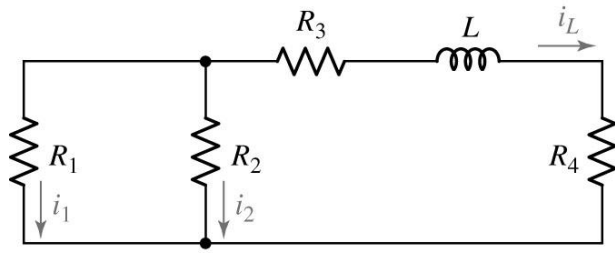
Quiz

Find $v(0)$ and $v(2 \text{ ms})$



50 V; 18.39 V

General RL circuits



$$R_{eq} = R_3 + R_4 + \frac{R_1 R_2}{R_1 + R_2}$$

If there is a number of resistors

$$\tau = \frac{L}{R_{eq}}$$

$$\tau = \frac{L_{eq}}{R_{eq}}$$

$$\tau = \frac{L_{eq}}{R_{TH}}$$

Let $i_L(0) \neq 0$ $i_L = i_L(0)e^{-t/\tau}$

Current division: $i_2 = -\frac{R_1}{R_1 + R_2} i_L(0) e^{-t/\tau}$

If we know the initial value of some other current $i_1(0^+)$: $i_2(0^+) R_2 = i_1(0^+) R_1$

$$i_L(0^+) = -[i_1(0^+) + i_2(0^+)] = -\frac{R_1 + R_2}{R_2} i_1(0^+)$$

$$i_2(0^+) = i_1(0^+) \frac{R_1}{R_2}$$

$$i_2 = i_1(0^+) \frac{R_1}{R_2} e^{-t/\tau}$$

Another way:

$$i_2 = A e^{-t/\tau}$$

$$i_2 = i_1(0^+) \frac{R_1}{R_2} e^{-t/\tau}$$

Example

Find i_1 and i_L for $t > 0$

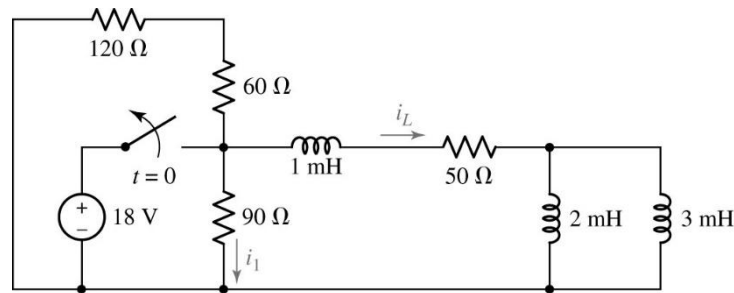
$$L_{eq} = 2.2 \text{ mH}$$

$$R_{eq} = 110 \Omega$$

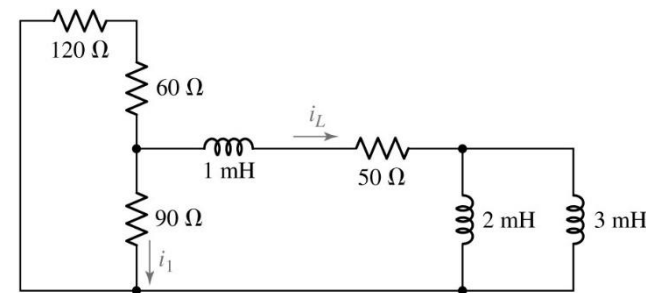
$$\tau = \frac{L_{eq}}{R_{eq}} = 20 \mu s$$

Response:

$$Ke^{-50000t}$$



(a)



(b)

$$i_L(0^-) = 18/50 = 360 \text{ mA} \quad i_L(0^-) = i_L(0^+)$$

$$i_L = \begin{cases} 360 \text{ mA}, & t < 0 \\ 360 e^{-50000t} \text{ mA}, & t \geq 0 \end{cases}$$

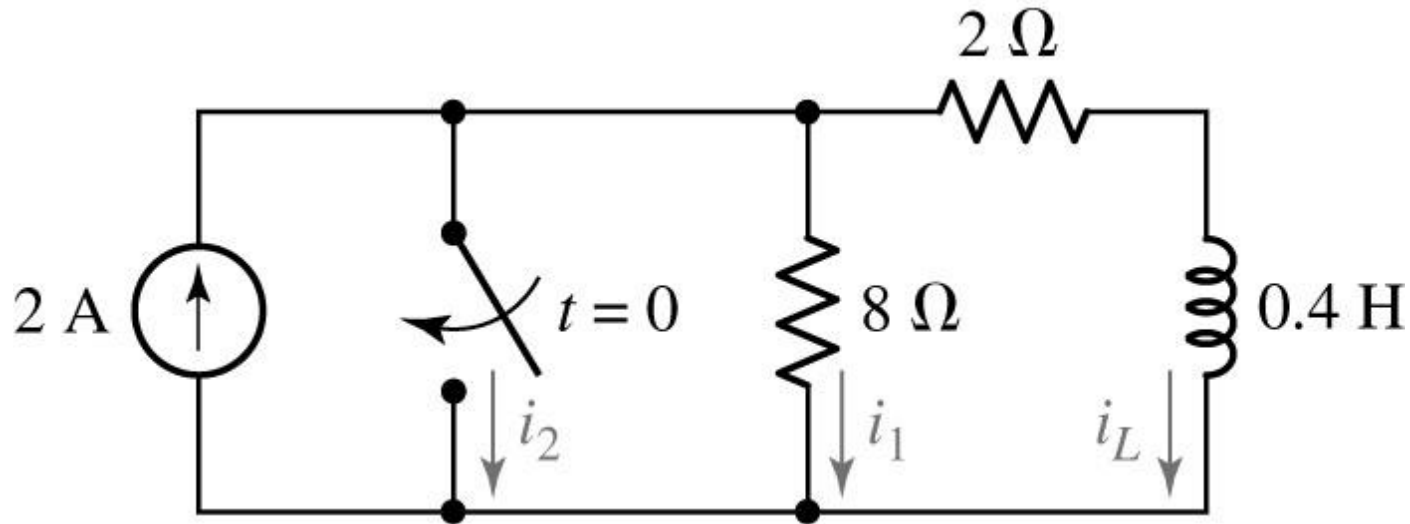
$$i_1(0^-) = 18/90 = 200 \text{ mA} \quad i_1(0^-) \neq i_1(0^+)$$

Current division:
$$i_1(0^+) = -i_L(0^+) \frac{120 + 60}{120 + 60 + 90} = -240 \text{ mA}$$

$$i_1 = \begin{cases} 200 \text{ mA}, & t < 0 \\ -240 e^{-50000t} \text{ mA}, & t \geq 0 \end{cases}$$

Practice

At $t = 0.15$ s find the values of i_L , i_1 , and i_2



Hint: Resistor 8 Ohm is short-circuited, and 2A source contributes its 100% current to i_2

0.756 A; 0; 1.244 A

Important note

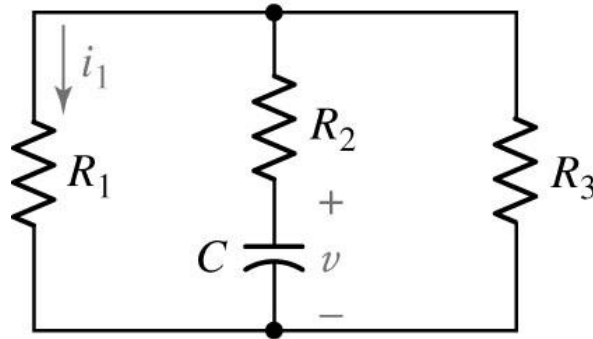
This approach valid if we can simplify circuit to 1 inductor and 1 resistor

Otherwise number of negative exponential terms will be equal to the number of inductors

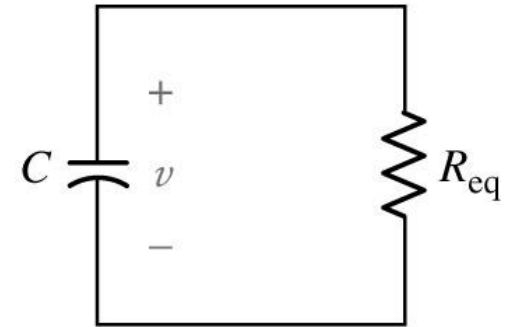
General RC circuits: EXAMPLE

$$\tau = R_{eq} C_{eq}$$

$$v = V_0 e^{-t/(R_{eq}C)}$$



(a)



(b)

Find $v(0^+)$ and $i_1(0^+)$ if $v(0^-) = V_0$

$$v(0^+) = v(0^-) = V_0$$

$$R_{eq} = R_2 + \frac{R_1 R_3}{R_1 + R_3}$$

Every current and voltage in resistive part of the circuit must have the form:

$$A e^{-t/(R_{eq}C)} \quad \text{So, for example} \quad i_1 = i_1(0^+) e^{-t/(R_{eq}C)}$$

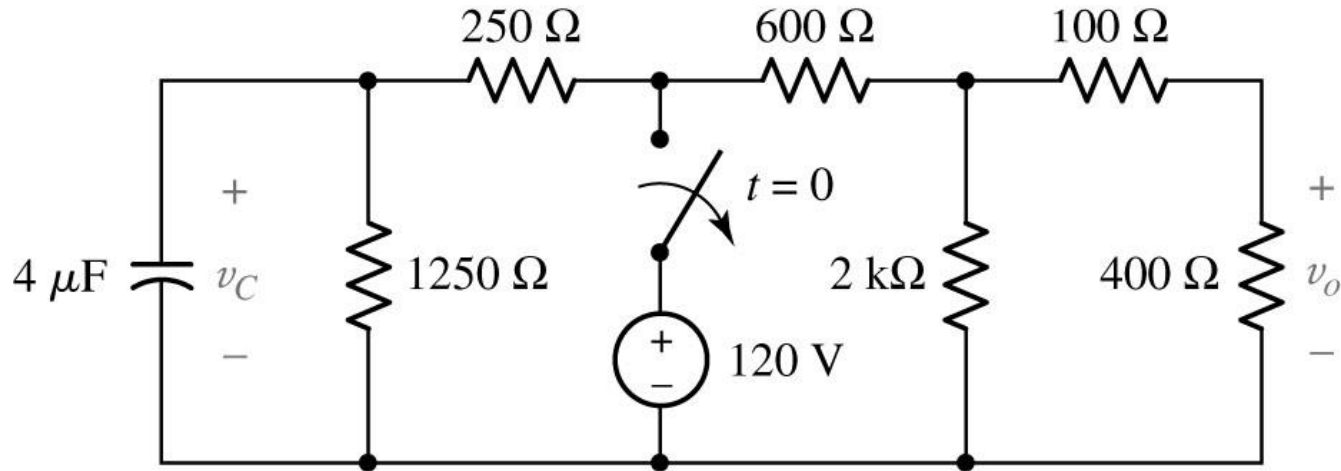
Since

$$v(0^+) = v(0^-) = V_0$$

$$i_1(0^+) = \frac{V_0}{R_2 + R_1 R_3 / (R_1 + R_3)} \frac{R_3}{R_1 + R_3}$$

Practice

Find values of v_C and v_o at t equal to (a) 0^- ; (b) 0^+ ; (c) 1.3 ms



(a) $100\ \text{V}$, $38.4\ \text{V}$; (b) $100\ \text{V}$, $25.6\ \text{V}$; (c) $59.5\ \text{V}$, $15.22\ \text{V}$

If dependent source is present

General approach

- Write an appropriate KCL or KVL equation
- Write necessary supporting equations
- Distill this down into a single differential equation
- Extract the characteristic equation to find the time constant

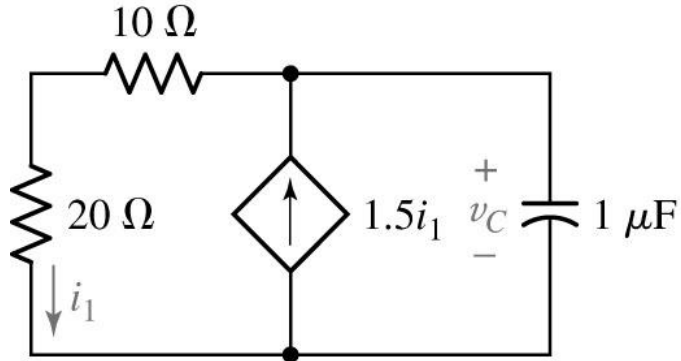
Alternative approach

- Find Thevenin equivalent resistance of the network connected to the capacitor or inductor
- Use the Thevenin equivalent in calculating the RC or RL time constant

If the dependent source is controlled by a voltage or current associated with the energy storage element, the Thevenin approach can not be used

Example

Find the voltage labeled v_C for $t > 0$ if $v_C(0^-) = 2 \text{ V}$



$$1.5i_1 = \frac{v_C}{30} + 10^{-6} \frac{dv_C}{dt}$$

$$\frac{dv_C}{dt} - \frac{1}{60 \cdot 10^{-6}} v_C = 0$$

$$s = \frac{1}{60 \cdot 10^{-6}}$$

$$i_1 = \frac{v_C}{30}$$

$$s - \frac{1}{60 \cdot 10^{-6}} = 0$$

$$A = v_C(0^+) = v_C(0^-) = 2 \text{ V}$$

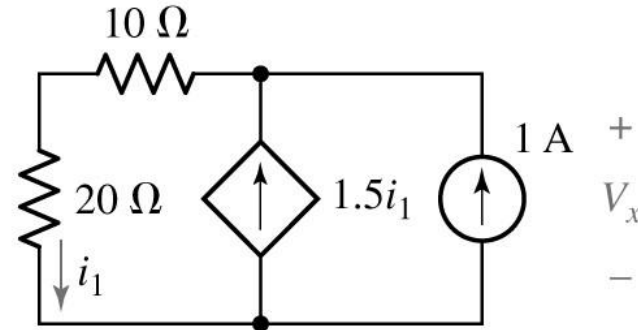
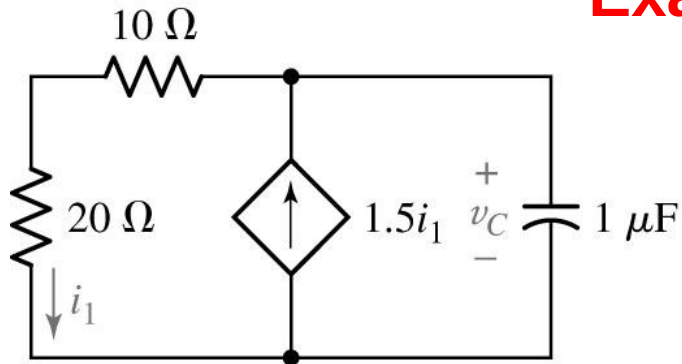
$$v_C(t) = 2e^{\frac{t}{60} \cdot 10^6} \text{ V}$$

UNSTABLE CIRCUIT: V_C growth exponentially with time;

One or more elements in the circuit will definitely fail

$$v_C(0) = 2V$$

Example



$$\frac{V_x}{30} = 1 + 1.5i_1 \quad i_1 = \frac{V_x}{30}$$

$$V_x = -60V$$

$$R_{TH} = -60 / 1 = -60 \Omega$$

$$\tau = -60(1 \cdot 10^{-6}) = -60 \mu\text{s}$$

$$v_C(t) = Ae^{t/60 \cdot 10^{-6}} V$$

$$A = v_C(0^+) = v_C(0^-) = 2V$$

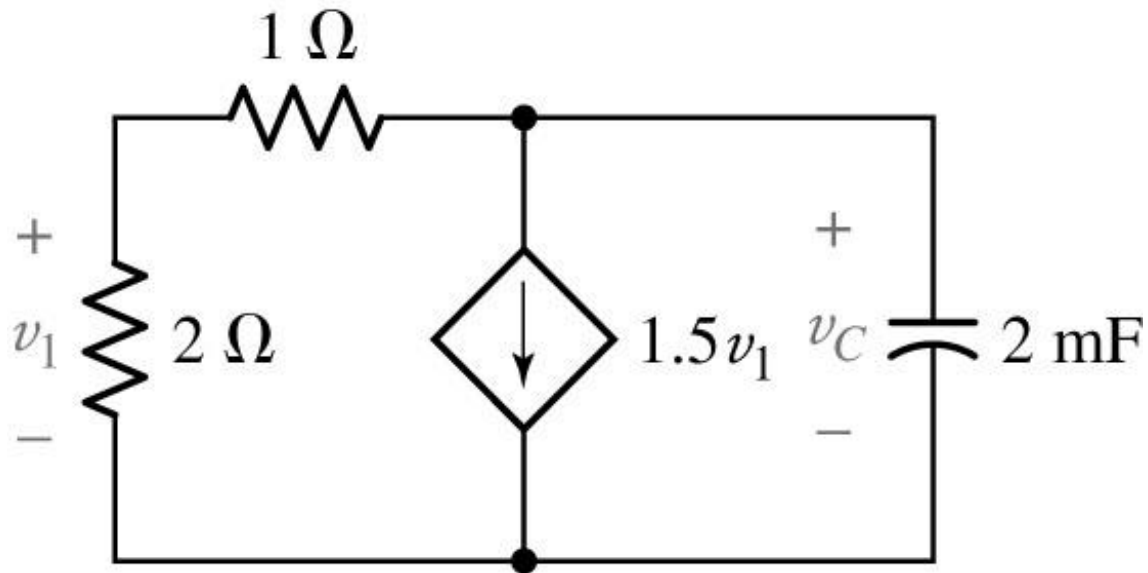
$$v_C(t) = 2e^{\frac{t}{60} \cdot 10^6} V$$

UNSTABLE CIRCUIT: V_C growth exponentially with time;

One or more elements in the circuit will definitely fail

Practice

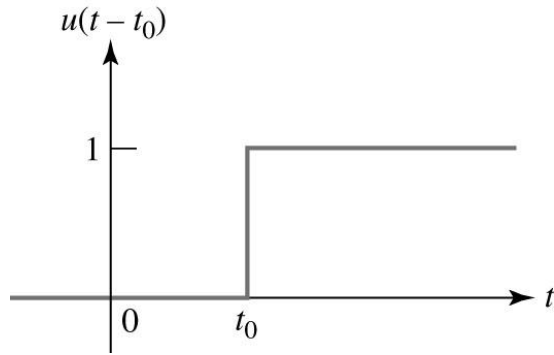
Determine the voltage $v_C(t)$ for $t > 0$ if $v_C(0^-) = 11$ V. Is the circuit stable?



$11\exp(-2\cdot 10^3 t/3)$ V at $t > 0$; Yes. Decays with time

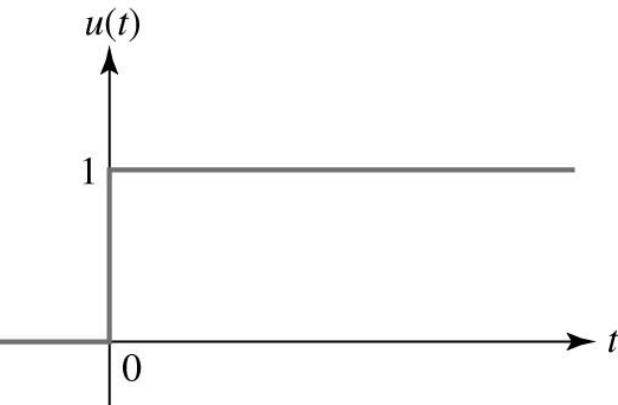
The Unit-Step Function

Energy sources suddenly applied to circuit



$$u(t - t_0) = \begin{cases} 0 & t < t_0 \\ 1 & t > t_0 \end{cases}$$

Special forcing functions – singularity functions which are discontinuous or have discontinuous derivatives: (unit-step function or unit-impulse function)



$$t_0 = 0$$

$$u(t) = \begin{cases} 0 & t < 0 \\ 1 & t > 0 \end{cases}$$

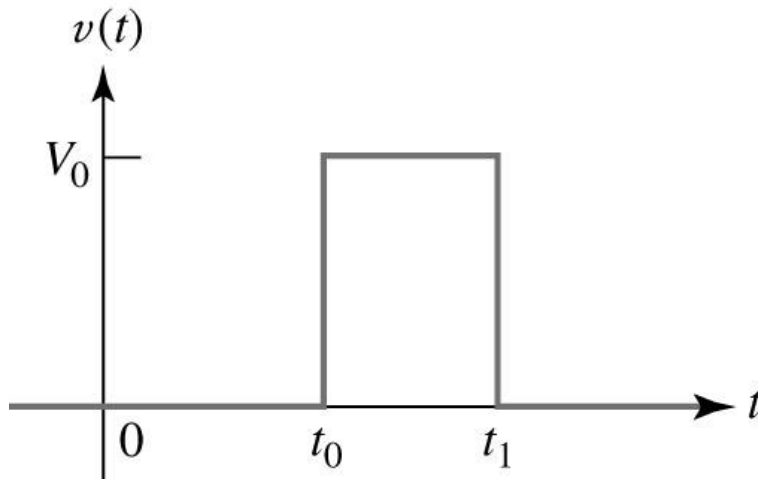
Unit-step function is dimensionless. If voltage should be represented, we need to multiply $u(t - t_0)$ by some constant voltage.

$$v(t) = 5u(t - t_0) \text{ for 5 V}$$

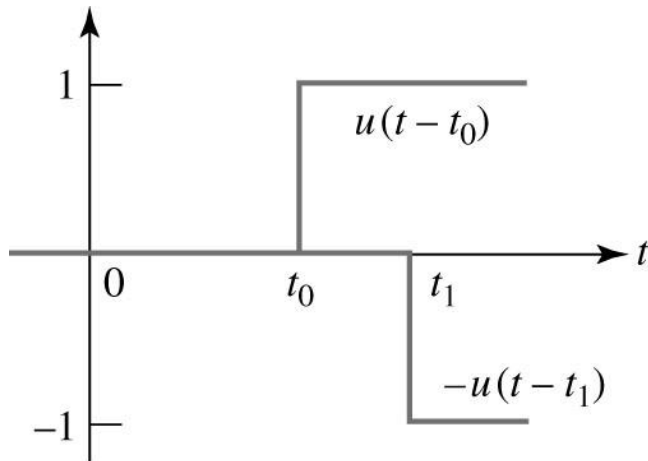
If we have a source $V_m \sin \omega t$ suddenly connected to a network at $t = t_0$

$$v(t) = V_m u(t - t_0) \sin \omega t$$

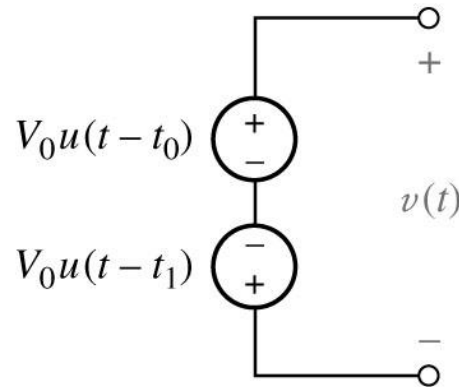
The Rectangular Pulse Function



$$v(t) = \begin{cases} 0 & t < t_0 \\ V_0 & t_0 < t < t_1 \\ 0 & t > t_1 \end{cases}$$



(a)



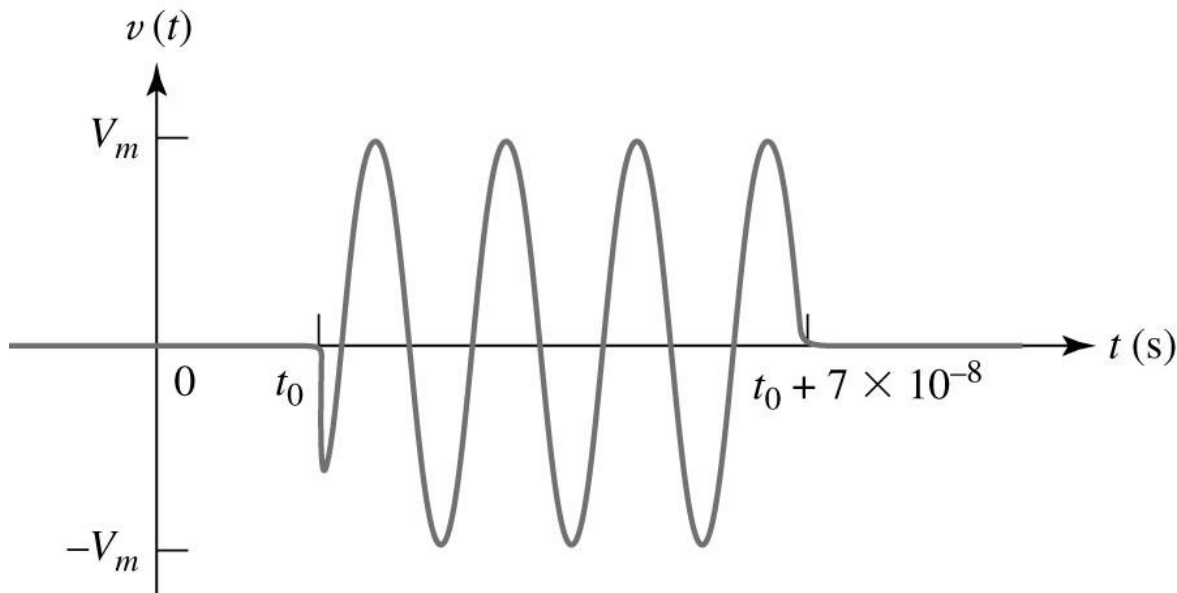
(b)

$$V_0 u(t - t_0) - V_0 u(t - t_1)$$

Example

If we wish to represent one burst of energy from the transmitter for a radio-controlled car operating at 47MHz (295 Mrads/s), we may turn the sinusoidal source off 70 ns later by a second unit-step forcing function

$$v(t) = V_m [u(t - t_0) - u(t - t_0 - 7 \times 10^{-8})] \sin(295 \times 10^6 t)$$



Practice

Evaluate each of the following at $t = 0.8$:

a) $3u(t) - 2u(-t) + 0.8u(1-t)$

b) $[4u(t)]u(-t)$

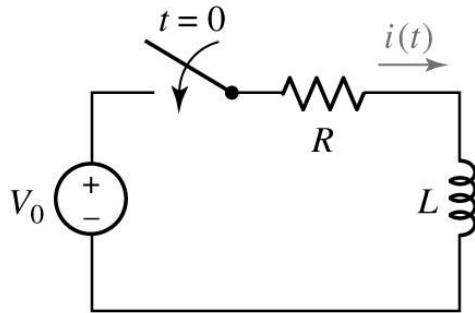
c) $2u(t)\sin\pi t$

$u(x)$ will evaluate to 1 if x is positive, and zero otherwise, whatever the expression you have in the place of x .

So, for $u(1-t)$, we need to see what values of t keep $(1-t)$ positive:

$$1-t > 0 \Rightarrow t < 1$$

Driven RL circuits

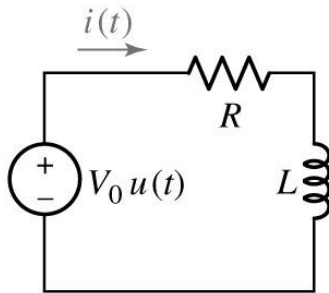


(a)

$$Ri + L \frac{di}{dt} = V_0 u(t)$$

$$i(t) = 0 \quad t < 0$$

$$Ri + L \frac{di}{dt} = V_0 \quad t > 0$$



(b)

$$\frac{L di}{V_0 - Ri} = dt$$

$$-\frac{L}{R} \ln(V_0 - Ri) = t + k$$

$$i(0^-) = i(0^+) = 0$$

$$-\frac{L}{R} \ln(V_0) = k$$

$$-\frac{L}{R} [\ln(V_0 - Ri) - \ln V_0] = t$$

$$\frac{(V_0 - Ri)}{V_0} = e^{-Rt/L}$$

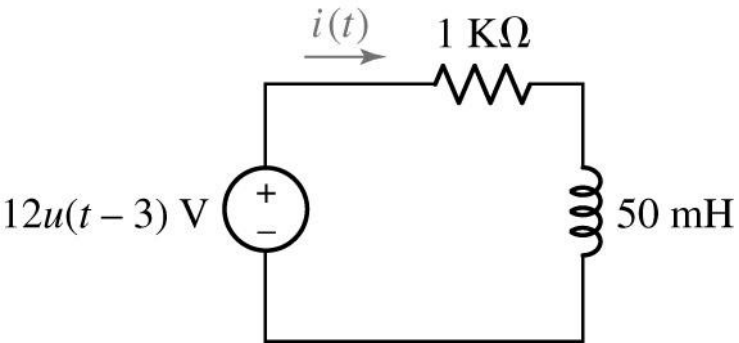
$$i = \frac{V_0}{R} - \frac{V_0}{R} e^{-Rt/L} \quad t > 0$$

$$i = \left(\frac{V_0}{R} - \frac{V_0}{R} e^{-Rt/L} \right) u(t)$$

The complete response is composed of **forced response** + **natural response**

Example

Find $i(t)$ for $t = \infty$; 3^- , 3^+ , and $100 \mu s$ after the source changes value



$$i(\infty) = \frac{12}{1000} = 12 \text{ mA}$$

$$u(t-3) = 0 \quad t < 3 \Rightarrow i(3^-) = 0$$

$$i(3^-) = i(3^+) = 0$$

$$i(t') = \left(\frac{V_0}{R} - \frac{V_0}{R} e^{-Rt'/L} \right) u(t') \quad t' = t - 3$$

$$i(t-3) = (12 - 12e^{-20000(t-3)})u(t-3) \quad \text{mA}$$

$$i(t) = (12 - 12e^{-20000(t-3)})u(t-3) \quad \text{mA}$$

$$i = 10.38 \text{ mA} \quad t = 3 \text{ s} + 100 \mu \text{s} = 3.0001 \text{ s}$$

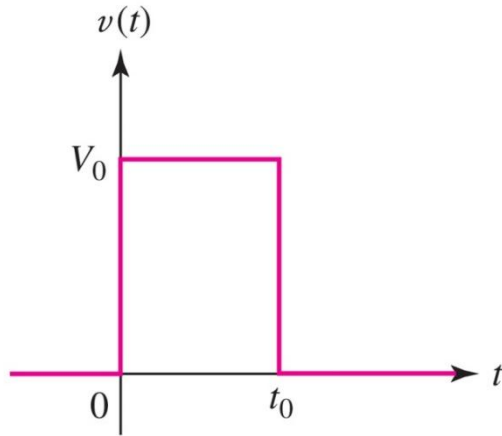
Practice

Find the current response in a simple series RL circuit when the forcing function is a rectangular voltage pulse of amplitude V_0 and duration t_0 .

$$\dot{i} = \dot{i}_1 + \dot{i}_2$$

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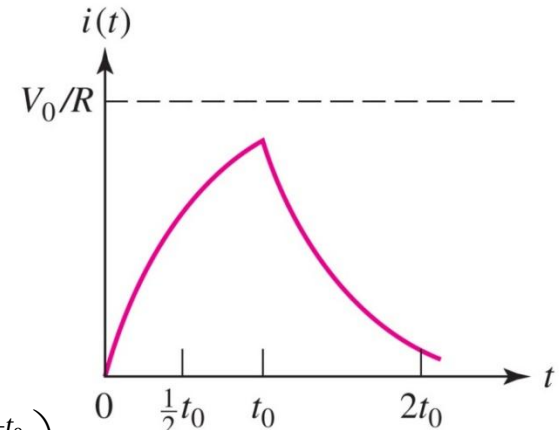


(a)

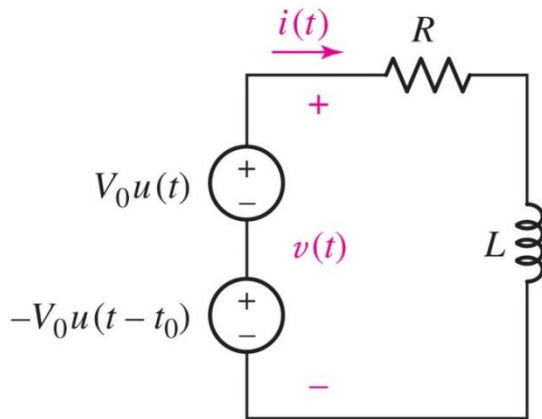
$$i = 0$$

$$i = \frac{V_0}{R} \left(1 - e^{-\frac{t}{\tau}} \right)$$

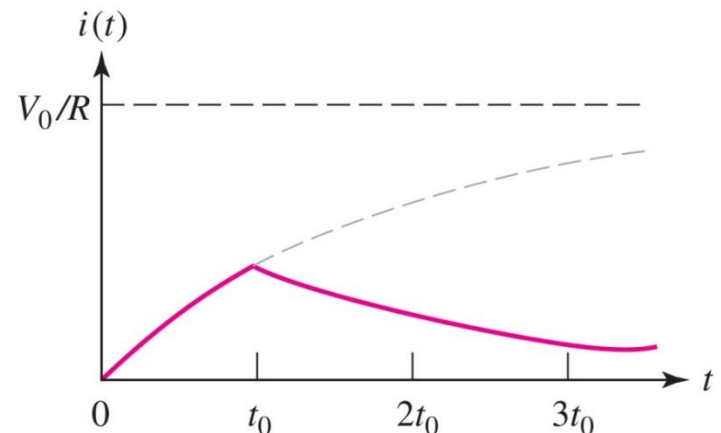
$$i = \frac{V_0}{R} \left(1 - e^{-\frac{t}{\tau}} \right) - \frac{V_0}{R} \left(1 - e^{-\frac{t-t_0}{\tau}} \right)$$



(a)



(b)



(b)

Natural and Forced Response

$$\frac{di}{dt} + Pi = Q(t) \quad / \text{ multiply by } e^{Pt}dt$$

$$i(t) = e^{-Pt} \int Q(t) e^{Pt} dt + Ae^{-Pt}$$



Forced response



Natural response

$$i_n = Ae^{-Pt}$$

DC source ($Q=\text{const}$):

Forced response (steady-state response, particular solution)

$$i_f = \frac{Q}{P}$$

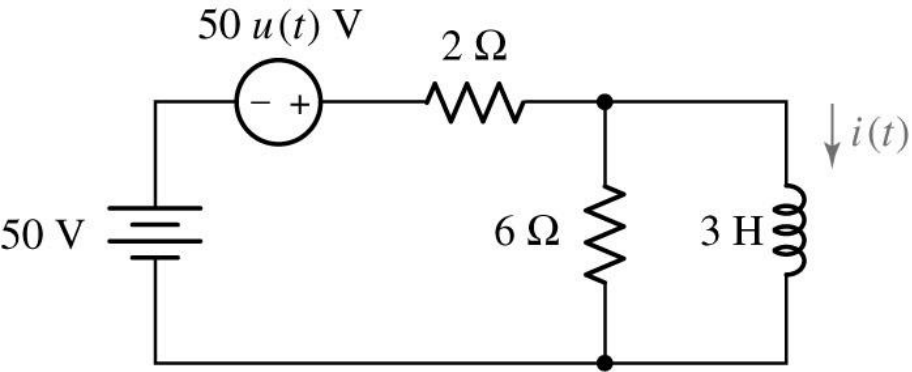
Complete response

$$i(t) = \frac{Q}{P} + Ae^{-Pt}$$

$$i = i_n + i_f$$

Example

Determine $i(t)$ for all values of time



$$\tau = \frac{L}{R_{th}} = \frac{3}{1.5} = 2 \text{ s}$$

$$i = i_n + i_f$$

$$i_n = K e^{-t/2} \quad t > 0$$

$$i_f = \frac{V_0}{R} = \frac{100}{2} = 50 \text{ A}$$

$$i = 50 + K e^{-t/2} \quad t > 0$$

$$i = 25 \text{ A} \quad t < 0 \quad 25 = 50 + K \quad t = 0 \quad \longrightarrow \quad K = -25$$

$$i = 50 - 25 e^{-t/2} \quad t > 0$$

$$f(t) = f(\infty) + A e^{-t/2}$$

$$i = 50 + (25 - 50) e^{-t/2}$$

$$f(t) = f(\infty) + (f(0^+) - f(\infty)) e^{-t/2}$$

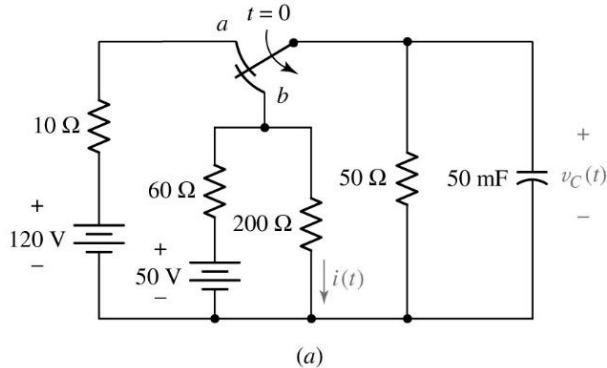
Driven RL Circuits Summary

1. With all independent sources zeroed out, simplify the circuit to determine R_{eq} , L_{eq} , and the time constant $\tau = L_{eq}/R_{eq}$.
2. Viewing L_{eq} as a short circuit, use dc analysis methods to find $i_L(0^-)$, the inductor current just prior to the discontinuity.
3. Again viewing L_{eq} as a short circuit, use dc analysis methods to find the forced response. This is the value approached by $f(t)$ as $t \rightarrow \infty$; we represent it by $f(\infty)$.
4. Write the total response as the sum of the forced and natural responses: $f(t) = f(\infty) + Ae^{-t/\tau}$.
5. Find $f(0^+)$ by using the condition that $i_L(0^+) = i_L(0^-)$. If desired, L_{eq} may be replaced by a current source $i_L(0^+)$ [an open circuit if $i_L(0^+) = 0$] for this calculation. With the exception of inductor currents (and capacitor voltages), other currents and voltages in the circuit may change abruptly.
6. $f(0^+) = f(\infty) + A$ and $f(t) = f(\infty) + [f(0^+) - f(\infty)] e^{-t/\tau}$, or total response = final value + (initial value - final value) $e^{-t/\tau}$.

Driven RC Circuits - EXAMPLE

Find the capacitor voltage $v_C(t)$ and the current $i(t)$ in the $200\ \Omega$ resistor for all times

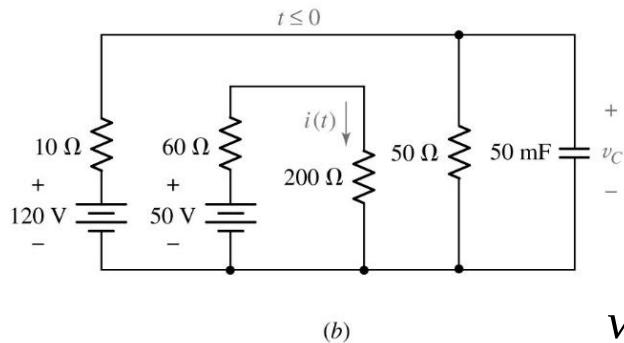
Switch at position a): $t < 0$



$$v_C(0^-) = v_C(0^+) = \frac{50}{50 + 10}(120) = 100V$$

Switch at position b): $t > 0$

$$v_C = v_{Cf} + v_{Cn}$$

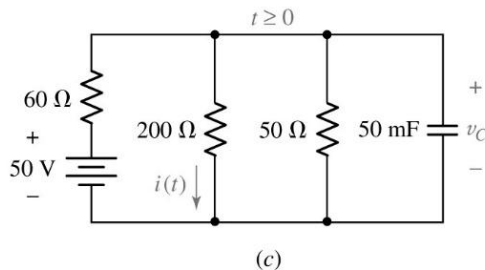


$$R_{eq} = \frac{1}{\frac{1}{200} + \frac{1}{50} + \frac{1}{60}} = 24\Omega$$

$$v_{Cn} = Ae^{-t/1.2}$$

$$v_{Cf} = 50\left(\frac{200 \parallel 50}{60 + 200 \parallel 50}\right) = 20V$$

$$v_C = 20 + Ae^{-t/1.2}$$

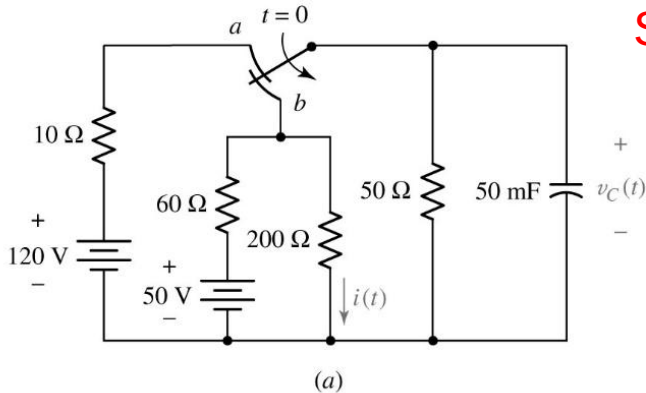


$$100 = 20 + A$$

$$v_C = 20 + 80e^{-t/1.2}$$

Driven RC Circuits - EXAMPLE

Find the capacitor voltage $v_C(t)$ and the current $i(t)$ in the $200\ \Omega$ resistor for all times

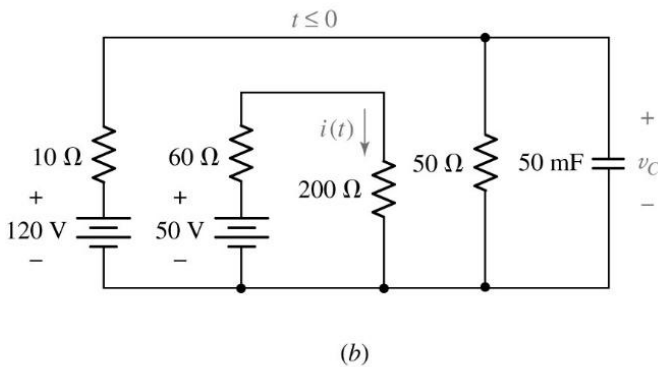


Switch at position a): $t < 0$ $i = 50 / 260 = 192.3 \text{ mA}$

Switch at position b): $t > 0$

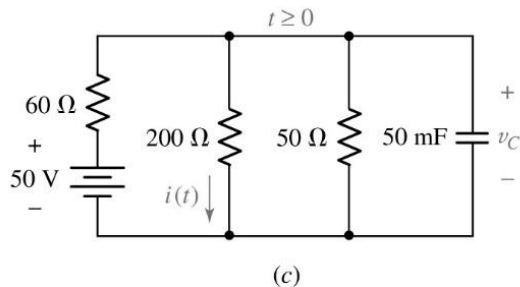
$$i_f = \frac{50}{60 + (50)(200)/(50 + 200)} \left(\frac{50}{50 + 200} \right) = 0.1 \text{ A}$$

$$i_n = Ae^{-t/1.2} \quad i = 0.1 + Ae^{-t/1.2}$$



$$i(0^+) = v_C(0^+) / 200 = 0.5 \text{ A}$$

$$0.5 = 0.1 + A \quad i(t) = 0.1 + 0.4e^{-t/1.2} \quad t > 0$$



At any time t

$$i(t) = 0.1923u(-t) + (0.1 + 0.4e^{-t/1.2})u(t)$$

Driven RC Circuits Summary

1. With all independent sources zeroed out, simplify the circuit to determine R_{eq} , C_{eq} , and the time constant $\tau = R_{eq}C_{eq}$.
2. Viewing C_{eq} as an open circuit, use dc analysis methods to find $v_C(0^-)$, the capacitor voltage just prior to the discontinuity.
3. Again viewing C_{eq} as an open circuit, use dc analysis methods to find the forced response. This is the value approached by $f(t)$ as $t \rightarrow \infty$; we represent it by $f(\infty)$.
4. Write the total response as the sum of the forced and natural responses: $f(t) = f(\infty) + Ae^{-t/\tau}$.
5. Find $f(0^+)$ by using the condition that $v_C(0^+) = v_C(0^-)$. If desired, C_{eq} may be replaced by a voltage source $v_C(0^+)$ [a short circuit if $v_C(0^+) = 0$] for this calculation. With the exception of capacitor voltages (and inductor currents), other voltages and currents in the circuit may change abruptly.
6. $f(0^+) = f(\infty) + A$ and $f(t) = f(\infty) + [f(0^+) - f(\infty)]e^{-t/\tau}$,
or total response = final value + (initial value - final value) $e^{-t/\tau}$.